

An Efficient Regenerated Cross-Modal Hashing: Improving Existing Hash Codes With the Arbitrary Length

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Abstract—In recent years, numerous hashing techniques have been developed to boost efficient cross-modal retrieval. Once a retrieval model is deployed, the hash code length is fixed to achieve optimal performance. To address different retrieval scenarios while maintaining retrieval accuracy, a common approach is to redesign and retrain the original model with different hash code length. However, this retraining process can increase the training load and may lead to worse results. To tackle these challenges, we present Regenerated Cross-Modal Hashing (RCMH), a novel cross-modal hashing framework designed to improve the quality of existing hash codes and convert them to arbitrary lengths with high efficiency. First, we clip or pad the existing hash codes to initialize them with the target length, under the supervision of the similarity matrix generated by the augmented label information. Second, we introduce a linear-nonlinear competitive reconstruction approach to reduce the semantic gaps and further capture the deeper relationships from linear image features and nonlinear text features. In this way, each pair of samples is compared and selected to obtain reconstructed binary codes that can preserve the modality-specific properties. Finally, to reduce the training costs caused by iterations of variables, the regenerate hashing term is utilized to regenerate final hash codes with the reconstructed binary codes while preserving the information from the existing hash codes without iterative optimization. Notably, RCMH can be integrated with existing state-of-the-art (SOTA) methods with robustness, helping them to adjust the hash code length and achieve better retrieval performance.

Index Terms—Cross-modal retrieval, hashing, augmented label information, linear-nonlinear reconstruction, regenerated framework.

I. INTRODUCTION

IN RECENT years, large multimodal AI models have achieved significant progress in diverse fields [1], [2], [3]. However, smaller, more efficient semantic retrieval models are urgently needed for computationally resource-constrained devices. Hashing technology can compact complex instances into Hamming space, accelerating retrieval and reducing computational overhead [4], [5], [6], [7], [8]. Consequently, cross-modal retrieval is often integrated with hashing techniques to enable rapid, accurate retrieval across different modalities [9], [10], [11], [12], [13]. To minimize the semantic gap between modalities, cross-modal hashing methods typically learn hash functions that map high-dimensional data from various modalities into a common semantic subspace [14], [15], [16], [17], [18]. Matrix factorization-based methods are among the first to address this challenge. The collective matrix factorization (CMF) framework attempts to linearly decompose data from different modalities into a common subspace [19], [20], [21], [22], [23]. Despite its efficiency, the linear CMF framework has limitations that lead to suboptimal retrieval outcomes. For example, it struggles to capture and quantify pairwise correlations between samples. To address these limitations, the similarity matrix factorization (SMF) framework was introduced. SMF preserves pairwise similarity information within hash codes by constructing a pairwise similarity matrix using label information, which is then decomposed to generate hash codes that preserve label pairwise relationships [24], [25], [26], [27]. This approach helps in capturing the semantic relationships between different data instances. In practice, combining CMF and SMF frameworks has proven effective in achieving superior retrieval results and enhancing the robustness of retrieval systems [28], [29].

Although these methods can achieve satisfactory results when combined in most scenarios, they encounter significant challenges once their models are determined and hash codes are generated. The common way to change the hash code length and improve retrieval results is to retrain and redefine their models, as illustrated in Fig. 1(a). Despite modifications to various modules aimed at boosting retrieval efficiency, it remains challenging to identify which specific modules effectively contribute to this

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The code is available at <https://github.com/kaihang-jiang/RCMH.git>.

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TABLE I
NOTATION TABLE

Symbol and Dimensions	Description
m, c, n	Numbers of modalities, categories and samples, respectively
$\mathbf{X}_m \in \mathbb{R}^{d_m \times n}, \mathbf{L} \in \mathbb{R}^{c \times n}$	Feature matrix of data in the t -th modality, label matrix
$\mathbf{U} \in \mathbb{R}^{r \times n}, \mathbf{B} \in \mathbb{R}^{f \times n}$	Existing hash codes matrix, final hash codes matrix
r, f	Bit-length of the existing hash codes, final hash codes
$\mathbf{E} \in \mathbb{R}^{c \times n}, \mathbf{F} \in \mathbb{R}^{c \times n}$	Augmented label matrix, negative label matrix
$\mathbf{M} \in \mathbb{R}^{c \times n}$	Weighted matrix
$\hat{\mathbf{X}}_m \in \mathbb{R}^{q \times n}$	Feature matrix of data after normalization
$\mathbf{W}_m \in \mathbb{R}^{c \times d_m}$	Orthogonal projection in the t -th modality
$\mathbf{R} \in \mathbb{R}^{r \times f}, \mathbf{V} \in \mathbb{R}^{r \times n}$	Bit-conversion matrix, Real-valued matrix
$\mathbf{Q} \in \mathbb{R}^{r \times d_1}, \mathbf{D} \in \mathbb{R}^{d_1 \times d_1}$	Projection matrix, diagonal matrix
$\mathbf{Y} \in \mathbb{R}^{d_1 \times n}, \mathbf{H}_2 \in \mathbb{R}^{d_2 \times n}$	Matrix after SVD, nonlinear projection matrix
$\mathbf{Z}_1 \in \mathbb{R}^{r \times n}, \mathbf{Z}_2 \in \mathbb{R}^{r \times n}$	Reconstruction image codes, text codes
$\mathbf{Z} \in \mathbb{R}^{r \times n}, \mathbf{A} \in \mathbb{R}^{n \times n}$	Reconstructed binary codes, competitive values matrix

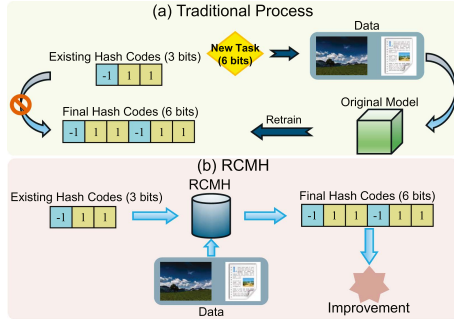


Fig. 1. The differences between (a) traditional process and (b) RCMH.

enhancement with high efficiency. Consequently, modifying the model and retraining existing hashing methods to obtain different hash code lengths and superior retrieval results often proves to be a costly and low-yield endeavor. This process typically results in extended training times and suboptimal retrieval performance. To the best of our knowledge, few studies have developed a scalable framework for cross-modal hashing that can directly adjust the length of existing hash codes in deployed models while further enhancing performance. Creating such a framework as a module within existing methods, capable of altering the hash code bit length and achieving improvements with minimal additional training cost, is both appealing and practical. This approach would allow existing retrieval methods to be more effective and efficient, enabling their learned hash codes to be directly converted into different bit lengths, which can help handle different retrieval scenarios, as illustrated in Fig. 1(b).

The design and implementation of a scalable module should focus on several common shortcomings of the existing methods to enhance their efficiency and effectiveness in bit length conversion. Firstly, traditional label enhancement techniques always learn enhanced label distributions from images [30], [31], [32]. But the label distributions learned from images by complex algorithms do not match the text, which is unsuitable in multi-modal tasks. Moreover, the label distributions are always represented by the Softmax function; thus, the '0' element in labels is given a positive value, which destroys the label structure itself. Secondly, to narrow the semantic gaps between different modalities, the previous methods always project features of different modalities into the unified subspace linearly, which might ignore the modality-specific properties. For example,

image features are generally continuous and linear, while text features are discrete and nonlinear. Moreover, how to compare and distinguish the properties in each pair of samples remains a challenge to further capture the pairwise deeper features. Finally, the traditional methods spend too much time and memory space in the iterative solution of the variables, so that retraining may lead to long training time and high memory occupancy.

To overcome these limitations, we introduce Regenerated Cross-Modal Hashing (RCMH), an innovative and efficient cross-modal hashing framework. It leverages existing hash codes from baselines as input features to enhance retrieval results with arbitrary bit lengths. The flowchart of the proposed RCMH is illustrated in Fig. 2. To improve label quality, RCMH generates augmented labels by preserving modality similarity and negative information while maintaining the label structure. These augmented labels are used to construct a label similarity matrix, which is subsequently factorized to generate a real-value space matrix and initial hash codes of arbitrary bit length. The initial hash codes are constrained to approximate the existing hash codes through an orthogonal projection, ensuring information from the existing codes is preserved. To retain modality-specific characteristics, RCMH introduces a novel competitive reconstruction strategy. To retain the modality-specific properties, we propose a novel competitive reconstruction strategy that extracts linear features from image data with linear factorization and extracts nonlinear features from text data with nonlinear projections. Then these features of each image-text pair are compared to obtain the reconstructed binary space, which distinguishes the linear and nonlinear features between each sample to preserve the pairwise modality-specific properties. Finally, a regenerated hashing term is employed to further refine the reconstructed binary space and generate the final hash codes. Since this term is based on the existing hash codes, it avoids the need for iterative optimization and significantly reduces training time and computational costs. The contributions of RCMH can be highlighted as:

- To the best of our knowledge, we are the first to propose a framework that integrates a highly scalable enhancement module for arbitrary bit length conversion without iterative optimizations. This framework not only boosts retrieval performance but also significantly reduces training costs when applied to existing methods. Its robustness enables the existing methods with poor performance to achieve satisfactory retrieval results.

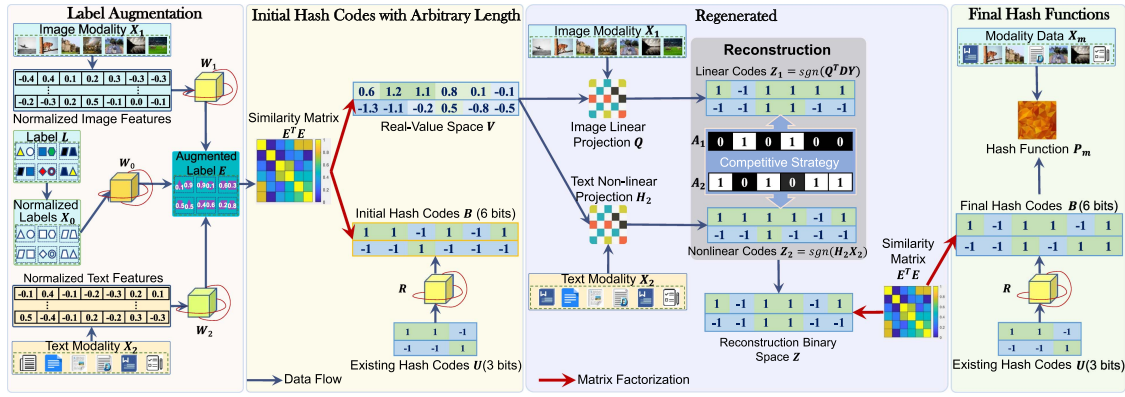


Fig. 2. The basic flow of RCMH. Initially, we perform label augmentation to construct the similarity matrix. Subsequently, we factorize the similarity matrix to obtain a real-value space and initial hash codes. Within the RCMH framework, image linear projection extracts linear features from images, and text nonlinear projection captures nonlinear features from text data. These features from different modalities are then compared through a competitive strategy to enhance the reconstructed binary space. The final hash codes are generated by integrating the reconstructed binary space while preserving information from the existing hash codes. Ultimately, hash functions are learned from these final hash codes.

- Unlike traditional hashing methods that factorize features from different modalities into a common space to eliminate semantic gaps, RCMH introduces a novel linear-nonlinear competitive strategy. This strategy aims to preserve modality-specific properties within each pair of samples. By dynamically comparing and selecting optimal features, it captures deeper relationships from image-text features and incorporates them into the reconstructed binary codes. Through this approach, RCMH concludes that nonlinear text features play a more significant role than linear image features in retrieval tasks.
- Extensive experiments on four public datasets demonstrate the superiority of RCMH over SOTA shallow cross-modal methods and its ability to outperform SOTA deep cross-modal methods.

The rest of this paper is organized as follows: Section II briefs the development of works related to RCMH. Section III introduces the methodology of RCMH and its theoretical analysis. Section IV shows the experimental results and comparisons with other shallow and deep SOTA methods on four public benchmarks. Finally, the conclusions and some future works are presented in Section V.

II. RELATED WORK

In this section, we will briefly introduce the related works, including label enhancement methods, collective matrix factorization hashing, similarity matrix factorization hashing and deep cross-modal hashing.

A. Label Enhancement Method

Manual labeling typically assigns binary values (0, 1) to data in retrieval tasks, but this method often fails to capture the detailed information in multi-label datasets [33], [34], [35]. For instance, there is an image labeled with “sea” and “human beings”, these labels alone do not provide information on the proportion of sea versus human beings. However, labels are frequently used for similarity comparisons in cross-modal retrieval tasks, and detailed label information can significantly

improve the construction of a high-quality similarity matrix, thereby enhancing retrieval performance. To capture such detailed labels, label enhancement methods have been proposed to extract more precise label distributions from images using manifold learning and low-rank techniques [36], [37]. For example, label distribution learning with label correlations via low-rank approximation (LDL-LCLR₂₀₁₉) [31] minimizes the information entropy between the actual label distribution and the predicted label distribution through low-rank approximation. Another approach, label enhancement for label distribution learning (GLLE₂₀₂₁) [38], incorporates K-nearest neighbors of original features and label similarities into the label distribution to generate enhanced labels. More recently, label distribution learning by exploiting label distribution manifold (LDL-LDM₂₀₂₃) [39] uses manifold learning to integrate global and local label correlations into label distributions, producing even more detailed labels. However, these label enhancement methods are not specifically designed for cross-modal tasks and cannot effectively handle cross-modal retrieval scenarios. Label distribution guided hashing (LADH₂₀₂₄) [40] learns a label distribution for each sample to weigh different labels via multi-modal features. While this approach effectively enriches semantic information, its reliance on Softmax functions transforms original zero elements (indicating label absence) into positive values. This can blur or distort the original label structure. Moreover, the resulting enhanced labels lack explicit negative information, which may widen the semantic gap between positive labels and binary hash codes (which contain −1 values). To overcome these limitations, RCMH introduces a novel label augmentation technique. This technique not only preserves the original label structure but also incorporates negative information, thereby reducing the semantic gap between the augmented labels and the target hash codes.

B. Collective Matrix Factorization Hashing

The CMF framework is always employed to explore the correlations between data from different modalities, premised on the assumption that data from these modalities can be factorized

into a shared subspace. Suppose $\mathbb{X}$ is the data of image modality, $\mathbb{Y}$ is the data of text modality, and the objective of the CMF-based method is shown as

$$\varphi(\mathbb{U}_1, \mathbb{U}_2, \mathbb{V}) = \beta_1 \|\mathbb{X} - \mathbb{U}_1 \mathbb{V}\|_F^2 + \beta_2 \|\mathbb{Y} - \mathbb{U}_2 \mathbb{V}\|_F^2 \quad (1)$$

where $\sum_{t=1}^2 \beta_t = 1$ is to balance the effectiveness of each modality, $\mathbb{V}$ is a common semantic subspace, $\mathbb{U}_1$ and $\mathbb{U}_2$ are decomposed factors for $\mathbb{X}$ and $\mathbb{Y}$, respectively. Collective matrix factorization hashing [19] made the first endeavor to utilize CMF in addressing the hashing learning problem for cross-modal retrieval. Owing to its remarkable performance, CMF has been extensively studied and expanded upon many literatures, such as adaptive label correlation based asymmetric discrete hashing (ALECH₂₀₂₃) [41], weakly-supervised enhanced semantic-aware hashing (WASH₂₀₂₃) [42], discrete online cross-modal hashing (DOCMH₂₀₂₄) [43] and three-stage semisupervised cross-modal hashing (TS3H₂₀₂₅) [44]. The common property of these supervised methods is that they decompose the modality features with label information into the common subspace.

C. Similarity Matrix Factorization Hashing

Different from CMF, the SMF framework focuses on preserving pairwise similarities between samples into hash codes. Given the similarity between the i -th and j -th samples as $\mathbb{S}_{ij}$, and their binary code representations as $\mathbf{b}_i$ and $\mathbf{b}_j$, respectively, SMF minimizes the following objective function:

$$\min_{\mathbf{b}_i, \mathbf{b}_j} \sum_{i=1}^N \sum_{j=1}^N \|\mathbb{S}_{ij} - \mathbf{b}_i \mathbf{b}_j^T\|_2^2 \quad \text{s.t. } \mathbf{b} \in \{-1, 1\}^r \quad (2)$$

where r is the length of binary codes. As a result, when $\mathbb{S}_{ij} = 1$, the binary code $\mathbf{b}_i$ be identical to $\mathbf{b}_j$, and when $\mathbb{S}_{ij} = -1$, $\mathbf{b}_i$ should be completely different from $\mathbf{b}_j$. Here, the similarity matrix $\mathbb{S}$ is always derived from label information $\mathbb{L}$ in supervised methods, i.e., $\mathbb{S} = \mathbb{L}^T \mathbb{L}$. A matrix tri-factorization hashing (MTFH₂₀₂₁) [45], fast cross-modal hashing (FCMH₂₀₂₂) [46], joint specifics and consistency hash learning (SCLCH₂₀₂₂) [47], robust online hashing with label semantic enhancement (ROHLSE₂₀₂₄) [48] and random online hashing (ROH₂₀₂₅) [49], all employ the SMF to learn hash codes and achieve SOTA performance. Although SMF is a well-established framework widely adopted by SOTA methods in cross-modal hashing, it is typically employed as a single-stage learning objective that directly maps input features and labels to hash codes. Consequently, such approaches lack the capacity to iteratively refine or semantically enhance the generated codes using multi-modal information. In contrast, RCMH leverages the SMF twice to regenerate high-quality binary codes that better align with the enriched semantics.

D. Deep Cross-Modal Retrieval

With the significant advancements in deep learning technology, deep cross-modal methods have emerged as an innovative method to achieve impressive performance in retrieval [50], [51], [52], [53]. Deep cross-modal hashing methods leverage deep neural networks to transform high-dimensional data from

diverse modalities into compact binary codes that preserve semantic similarity information [54], [55], [56]. Deep cross-modal hashing (DCMH₂₀₁₇) [57] pioneered this method by utilizing the Visual Geometry Group (VGG) network to extract image vectors, while text was represented as bag-of-word (BOW) vectors. Subsequent methods, such as deep cross-modal hashing with hashing functions (DCHUC₂₀₂₂) [58] and bit-aware semantic transformer hashing (BSTH₂₀₂₂) [59], also adopted similar feature extraction protocols as DCMH for their experimental evaluations. To construct the similarity matrix with intra-modal information, bidirectional reinforcement guided hashing (Bi-CMR₂₀₂₂) [60] proposed a bidirectional learning method to generate multi-label hash codes and refine the similarity matrix. Integrating multi-label contrastive learning with dual adversarial graph neural networks (I-GNNCON₂₀₂₃) [61] employed graph neural networks and a generative adversarial network to align the representation similarity with the label similarity and handle cross-modal retrieval tasks. To minimize subspace gaps, semantic channel hashing (SCH₂₀₂₄) [62] proposed classifying sample pairs to ensure the full utilization of the Hamming space and used a semantic channel to mitigate loss function oscillation. Recently, contrastive language-image pre-training (CLIP) is widely used to capture more complex information from images which obtain better retrieval results, such as a unified generative hashing (UMoED₂₀₂₅) [63], cross-modal hashing via diverse instances matching (DIMCH₂₀₂₅) [64] and deep discriminative boundary hashing (DDBH₂₀₂₅) [65]. Although these deep learning methods achieve improvements in cross-modal retrieval, shallow hashing methods can obtain better results through nonlinear features without complex neural networks.

III. METHODOLOGY

A. Notations

Given n training instances, the data matrix for the m -modality is denoted by $\mathbf{X}_m \in \mathbb{R}^{d_m \times n}$, where d_m is the dimensionality of the m -th modality, $m \in \{1, 2, \dots, o\}$ and o is the total number of modalities. The label matrix is represented as $\mathbf{L} \in \{0, 1\}^{c \times n}$, where c is the number of categories. The final hash codes are denoted by $\mathbf{B} \in \{-1, 1\}^{r \times n}$, where r is the length of the final hash codes. The objective of RCMH is to learn a scalable generalized framework that can gain improvements for existing hashing methods with their obtained hash codes $\mathbf{U} \in \{-1, 1\}^{f \times n}$, where f is the length of the existing hash codes. Table I illustrates all the symbols with their dimensions. In this way, the existing hash codes can cooperate with augmented labels and linear-nonlinear features to learn high-quality final hash codes $\mathbf{B}$ by RCMH.

B. Label Augmentation

Enhancing label quality has significant implications for improving retrieval tasks. To achieve this, label enhancement methods attempt to assign weights to labels from images. However, these methods are often designed for single-modal scenarios and involve highly complex computations [30], [31], [38], [39], [66]. A common property of label enhancement methods is their use of the Softmax function to diffuse enhanced labels across

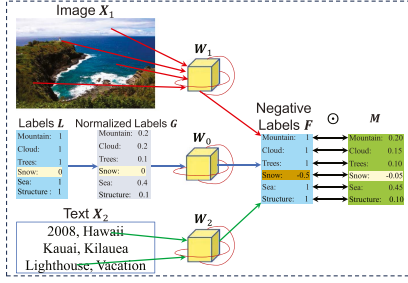


Fig. 3. The illustration of process in the label augmentation.

all categories. This approach may cause categories originally labeled as ‘0’ to have positive values. Such an issue can lead to distortion of the original label information and structure and introduce unwanted noise into the enhanced labels. To address these challenges, we propose a label augmentation method to obtain high-quality labels by preserving label structure and incorporating negative information into augmented labels. Furthermore, to optimize the weighted information for all modalities and embed modality similarity into the augmented labels $\mathbf{E}$, each sample $\mathbf{x}_m$ is normalized to eliminate the bias among different samples in the matrix $\mathbf{X}_m$ by $\hat{\mathbf{x}}_m = \frac{\mathbf{x}_m}{\|\mathbf{x}_m\|_2}$. Here, $\mathbf{x}$ and $\hat{\mathbf{x}}$ represent the columns of $\mathbf{X}$ and $\hat{\mathbf{X}}$, respectively. Similar, the label information $\mathbf{L}$ is normalized as $\mathbf{G}$. For conciseness, we define $\hat{\mathbf{X}}_0 = \mathbf{G}$ to preserve normalized label information into the augmented labels $\mathbf{E}$ as follows:

$$\min_{\mathbf{W}_m, \mathbf{M}} \sum_{m=0}^o \alpha_m \|\mathbf{W}_m \hat{\mathbf{X}}_m - \mathbf{E}\|_F^2 \quad (3)$$

where $\sum_{m=0}^o \alpha_m = 1$ is used to balance the effectiveness between the modality features and label information. To ensure equal contribution of each modality data $\hat{\mathbf{X}}_m$ ($m > 0$) to the augmented labels, we set $\alpha_1 = \alpha_2 = \dots = \alpha_m$ across all modality features. Once α_0 is determined, α_m can be calculated as $\alpha_m = \frac{1-\alpha_0}{o}$. As stated in *Proposition 1* of the **appendix** (available online), an orthogonal projection matrix can project all features into a common subspace while preserving local pairwise similarities. To better distinguish between positive and ‘0’ labels during optimization, we appropriately penalize the ‘0’ entries in the label matrix $\mathbf{L}$. Specifically, we replace all ‘0’ elements as negative parameter k and set ‘ $k = -0.5$ ’ in $\mathbf{L}$ to obtain the negative labels $\mathbf{F}$ according to experimental Section IV-J. To assign label weights without altering their positive or negative signs, the augmented labels $\mathbf{E}$ are redefined as $\mathbf{E} = \mathbf{M} \odot \mathbf{F}$, where $\mathbf{M} \in \mathbb{R}^{c \times n}$ is a positive weighted matrix. Overall, (3) can be further described as follows:

$$\min_{\mathbf{W}_m, \mathbf{M}} \sum_{m=0}^o \alpha_m \|\mathbf{W}_m \hat{\mathbf{X}}_m - \mathbf{M} \odot \mathbf{F}\|_F^2 \quad s.t. \mathbf{M} > 0 \quad (4)$$

By optimizing (4), the proposed label augmentation method can be effectively applied to multi-modal scenarios. Label augmentation preserves modality similarities and incorporates adaptive negative information into the augmented labels without altering their original structure, as illustrated in Fig. 3.

C. Initial Hash Codes With Arbitrary Length

Owing to the bit length difference $f \neq r$ and semantic discrepancy between the existing hash codes and the final hash codes, it is difficult to embed the existing hash codes into the final hash codes directly. To address these issues and produce high-quality hash codes, initializing the hash codes to match the bit length r of the final hash codes is crucial for handling bit length mismatches. Consequently, a straightforward SMF model is employed to initialize the final hash codes $\mathbf{B}$, i.e., $\|r\mathbf{G}^T\mathbf{G} - \mathbf{V}^T\mathbf{B}\|_F^2 + \xi\|\mathbf{V} - \mathbf{B}\|_F^2 + \mu\|\mathbf{B} - \mathbf{R}\mathbf{U}\|_F^2$. Here, the orthogonal matrix $\mathbf{R}$ is introduced to preserve local similarities and maintain the linear relationship between $\mathbf{U}$ and the final hash codes $\mathbf{B}$. The matrix $\mathbf{V} \in \mathbb{R}^{r \times n}$ is a real-valued matrix constrained to be balanced and pairwise uncorrelated [67]. According to *Proposition 2* of the **appendix** (available online), the above equation can be simplified as follows:

$$\min_{\mathbf{B}, \mathbf{V}} \sum_{m=1}^o \|r\mathbf{E}^T\mathbf{E} - \mathbf{B}^T\mathbf{V}\|_F^2 + \mu\|\mathbf{B} - \mathbf{R}\mathbf{U}\|_F^2 \quad (5)$$

s.t. $\mathbf{V}\mathbf{V}^T = n\mathbf{I}_r, \mathbf{V}\mathbf{1}_n = \mathbf{0}_r, \mathbf{R}^T\mathbf{R} = \mathbf{I}$

In this way, the final hash codes are initialized to the arbitrary bit length f while preserving the information from existing hash codes.

D. Linear-Nonlinear Competitive Reconstruction

The initial hash codes $\mathbf{B}$ obtained by (5) may only capture linear properties and exhibit inferior performance on large-scale complex datasets due to the inherited linear and nonlinear properties of data. For instance, after dimensionality reduction using methods like GIST and PCA, image features $\mathbf{X}_1$ tend to show continuous linear properties, whereas text features $\mathbf{X}_2$ present discrete nonlinearity through bag-of-words (BOW) vectors. At present, two-stage hashing methods commonly use linear regression to learn the hash function $\mathbf{H}_m$ from the hash codes via the equation $\min \|\mathbf{B} - \mathbf{H}_m\mathbf{X}_m\|_F^2$. Although efficient, this approach results in significant reconstruction errors between $\mathbf{H}_m$ and the final hash codes $\mathbf{B}$, and fails to distinguish the linear and nonlinear properties of images and texts. The reason is that the learned linear hash function $\mathbf{H}_m = \mathbf{B}\mathbf{X}_m^T(\mathbf{X}_m\mathbf{X}_m^T)^{-1}$ provides a biased projection from modality data $\mathbf{X}_m$. In real-world datasets, $\mathbf{X}_m$ is usually not a square matrix ($n \gg d_m$), leading to $\mathbf{H}_m\mathbf{X}_m = \mathbf{B}\mathbf{X}_m^T(\mathbf{X}_m\mathbf{X}_m^T)^{-1}\mathbf{X}_m \neq \mathbf{B}$ under $\mathbf{X}_m^T(\mathbf{X}_m\mathbf{X}_m^T)^{-1}\mathbf{X}_m \neq \mathbf{I}$. Consequently, the obtained linear hash functions $\mathbf{H}_m$ are not well-adapted to complex retrieval tasks. Furthermore, distinguishing linear and nonlinear features between cross-modal data is crucial for representation learning, enabling reconstructed features to possess a more comprehensive and accurate representation ability. To embed more comprehensive features into the final hash codes, we propose a novel linear-nonlinear competitive method as follows.

Linear Image Features: To exploit linear features from images, we assume that an orthogonal rotation matrix $\mathbf{Q}$ can embed image features into hash codes by minimizing $\|\mathbf{B} - \mathbf{Q}^T\mathbf{X}_1\|_F^2$. According to *Proposition 2* of the **appendix** (available online), $\mathbf{Q}$ projects the inter-similarity of image features into hash

codes $\mathbf{B}$. Replacing discrete hash codes $\mathbf{B}$ with continuous spaces $\mathbf{V}$ enhances linear decomposition reliability, yielding $\|\mathbf{V} - \mathbf{Q}^T \mathbf{X}_1\|_F^2$. Simplifying and preserving key information, we decompose $\mathbf{V}$ as $\mathbb{R} \mathbf{D} \mathbf{Y}^T$ via skinny singular value decomposition (SVD). For conciseness, $\mathbf{Q}^T \mathbb{R}$ is combined into an orthogonal variable $\mathbf{Q}^T$, since $\mathbb{R}^T \mathbf{Q} \mathbf{Q}^T \mathbb{R} = \mathbf{I} = \mathbf{Q} \mathbf{Q}^T$. This allows linear reconstruction of image features as follows:

$$\min_{\mathbf{Q} \mathbf{Q}^T = \mathbf{I}, \mathbf{D}} \|\mathbf{V} - \mathbf{Q}^T \mathbf{D} \mathbf{Y}^T\|_F^2 \quad (6)$$

To regulate and determine the optimal direction for each bit vector in the real-valued matrix $\mathbf{V}$, we propose recalculating $\mathbf{D}$ as a diagonal matrix: $\mathbf{D} = \text{diag}(d^{(1)}, \dots, d^{(d_1)})$. Let *mean* represent the mean of the $d^{(i)}$ values. Each $d^{(i)}$ is then recalculated using the following equation:

$$d^{(i)} = \begin{cases} 1, & d^{(i)} \geq \text{mean} \\ -1, & d^{(i)} < \text{mean} \end{cases} \quad (7)$$

The matrix $\mathbf{D}$ takes the direction of $\mathbf{V}$ into account, enabling the adjustment of low-similarity eigenvalues and their corresponding eigenvectors to the opposite direction. By minimizing (6), the reconstructed codes for image features can be obtained as: $\mathbf{Z}_1 = \text{sgn}(\mathbf{Q}^T \mathbf{D} \mathbf{Y}^T)$, where $\text{sgn}(x)$ is the sign function of x . In this way, the obtained linear codes $\mathbf{Z}_1$ can better approximate $\mathbf{B}$ while preserving the linear properties.

Nonlinear Text Features: To capture nonlinear features from the text modality while maintaining the same sign as the real-valued matrix $\mathbf{V}$, the hyperbolic tangent activation function $\tanh$ is employed to fit $\mathbf{X}_2$ with $\mathbf{V}$:

$$\min_{\mathbf{H}} \|\mathbf{V} - \tanh(\mathbf{H}_2 \mathbf{X}_2)\|_F^2 \quad (8)$$

where the real-value matrix $\mathbf{V}$ is superior to binary codes $\mathbf{B}$ according to *Proposition III* of the **appendix** (available online), since $\mathbf{V}$ contains more continuous information than the discrete information in $\mathbf{B}$. By optimizing (8), the obtained $\mathbf{H}_2$ can project the modality features into a broader subspace while preserving the properties of $\mathbf{V}$, and captures more nonlinear features than traditional kernelization [4], [68]. Therefore, the reconstructed codes of text features can be obtained as $\mathbf{Z}_2 = \text{sgn}(\mathbf{H}_2 \mathbf{X}_2)$, which can get closer to $\mathbf{B}$ and preserve nonlinear properties.

Pairwise Competitive Strategy: Although $\mathbf{Z}_1$ and $\mathbf{Z}_2$ can preserve linear and nonlinear features from images and text, respectively, there are still semantic gaps between the reconstructed codes from different modalities. To minimize reconstruction gaps while preserving modality-specific information, we establish a competitive strategy to generate reconstructed binary codes $\mathbf{Z}$ from different modalities as follows:

$$\mathbf{Z} = \text{sgn}(\mathbf{Z}_1 \mathbf{A} + \mathbf{Z}_2 (\mathbf{I} - \mathbf{A})) \quad (9)$$

The competitive values matrix $\mathbf{A} = \text{diag}(a^{(1)}, a^{(2)}, \dots, a^{(n)})$ is a diagonal matrix where the i -th element $a^{(i)} \in \{0, 1\}$. Here, $\mathbf{A}$ controls the impact of embedding $\mathbf{Z}_1$ and $\mathbf{Z}_2$ into $\mathbf{V}$. If $a^{(i)} = 1$, it means that the i -th sample in the linear image features $\mathbf{z}_1^{(i)}$ is more semantically similar to the real-value $\mathbf{v}^{(i)}$. In this case, $\mathbf{z}^{(i)}$ is composed exclusively of the image linear features $\mathbf{z}_1^{(i)}$ rather than text features $\mathbf{z}_2^{(i)}$. Conversely, when $a^{(i)} = 0$, $\mathbf{z}^{(i)}$ selects only the text nonlinear feature $\mathbf{x}_2^{(i)}$. To obtain the competitive values $\mathbf{A}$, we compare $\mathbf{z}_1^{(i)}$ and $\mathbf{z}_2^{(i)}$ with $\mathbf{v}^{(i)}$ to quickly calculate

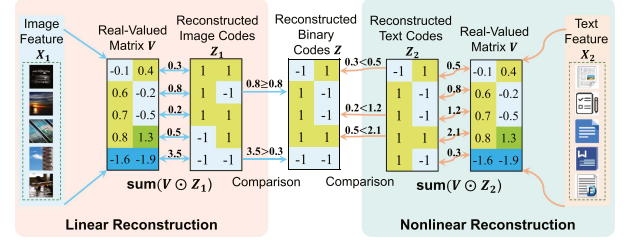


Fig. 4. Toy illustration of linear-nonlinear competitive reconstruction.

the results as:

$$a^{(i)} = \begin{cases} 1, & \text{sum}(\mathbf{z}_1^{(i)} \odot \mathbf{v}^{(i)}) \geq \text{sum}(\mathbf{z}_2^{(i)} \odot \mathbf{v}^{(i)}) \\ 0, & \text{sum}(\mathbf{z}_1^{(i)} \odot \mathbf{v}^{(i)}) < \text{sum}(\mathbf{z}_2^{(i)} \odot \mathbf{v}^{(i)}) \end{cases} \quad (10)$$

where $\text{sum}(\cdot)$ sums up all elements in the vector. Since $\mathbf{z}_1$ and $\mathbf{z}_2$ are binary codes, when the elements in $\mathbf{v}^{(i)}$ are consistent with $\mathbf{z}_1^{(i)}$, the similarity of $\text{sum}(\mathbf{z}_1^{(i)} \odot \mathbf{v}^{(i)})$ will be higher than $\text{sum}(\mathbf{z}_2^{(i)} \odot \mathbf{v}^{(i)})$. This strategy efficiently captures deeper features from different modalities by identifying semantically richer information and preserves modality-specific properties from each sample pair in the reconstructed binary space $\mathbf{Z}$. The details can be referred in Fig. 4. Notably, without deep learning feature extraction, nonlinear text features are found to be more important than linear image features, as detailed in the experimental Section IV-H.

E. Regenerated Final Hash Codes

RCMH reinterprets SMF from a ‘generative model’ to a ‘modular refinement and regeneration mechanism’. This design enables RCMH to significantly enhance the performance of existing hashing models and flexibly adapt their output to arbitrary code lengths at minimal computational cost and without re-optimizing all model parameters from scratch. To further embed the augmented label semantics into the final hash codes, we introduce SMF again to correct the reconstructed binary representations $\mathbf{Z}$ as follows:

$$\min_{\mathbf{B}, \mathbf{R}^T \mathbf{R} = \mathbf{I}} \|\mathbf{r} \mathbf{E}^T \mathbf{E} - \mathbf{B}^T \mathbf{Z}\|_F^2 + \mu \|\mathbf{B} - \mathbf{R} \mathbf{U}\|_F^2 \quad (11)$$

This regenerate term ensures that the final hash codes $\mathbf{B}$ align with the semantics of the augmented label information $\mathbf{E}$ while retaining the previously learned information $\mathbf{Z}$. The term $\|\mathbf{B} - \mathbf{R} \mathbf{U}\|_F^2$ helps preserve the existing hash codes $\mathbf{U}$ in the final hash codes $\mathbf{B}$. Since this term is based on the existing hash codes $\mathbf{U}$, it aids in quickly converging to optimal results without requiring multiple iterations.

F. Overall Objective Function

For a concise and precise representation of the objective function, we denote (4) as Loss_1 , (5) as Loss_2 , (6) as Loss_3 , (8) as Loss_4 , (9) as Loss_5 and (11) as Loss_6 . Therefore, the overall objective function Loss_{All} can be written as

$$\begin{aligned} \text{Loss}_{All} = & \min_{\mathbf{W}_m, \mathbf{M}, \mathbf{B}, \mathbf{V}, \mathbf{R}, \mathbf{Q}, \mathbf{D}, \mathbf{Z}, \mathbf{A}} \text{Loss}_1 + \text{Loss}_2 + \text{Loss}_3 \\ & + \text{Loss}_4 + \text{Loss}_5 + \text{Loss}_6 \\ \text{s.t. } & \mathbf{V} \mathbf{V}^T = n \mathbf{I}_r, \mathbf{V} \mathbf{1}_n = \mathbf{0}_r, \mathbf{R}^T \mathbf{R} = \mathbf{I}, \mathbf{Q} \mathbf{Q}^T = \mathbf{I} \end{aligned} \quad (12)$$

G. Optimization

RCMH adopts a sequential approach to update variables without the need for iterative optimization. The steps to update variables in sequence are as follows:

Step 1: To update $\mathbf{W}_m$, we fix variables $\mathbf{M}$, $\mathbf{R}$, $\mathbf{V}$, $\mathbf{B}$, $\mathbf{H}_m$, $\mathbf{D}$ and $\mathbf{Q}$. Then $Loss_1$ in (12) can be simplified as following:

$$\begin{aligned} \min_{\mathbf{W}_m} \|\mathbf{W}_m \hat{\mathbf{X}}_m - \mathbf{M} \odot \mathbf{F}\|_F^2 \\ = \max_{\mathbf{W}_m} Tr(\mathbf{W}_m \hat{\mathbf{X}}_m (\mathbf{M} \odot \mathbf{F})^T) \end{aligned} \quad (13)$$

Let $[\mathbf{U}, \phi, \mathbf{V}] = SVD(\mathbf{W}_m \hat{\mathbf{X}}_m (\mathbf{M} \odot \mathbf{F})^T)$, the solution of $\mathbf{W}_m$ can be obtained as

$$\mathbf{W}_m = \mathbf{U}\mathbf{V}^T \quad (14)$$

Step 2: Update label weighted matrix $\mathbf{M}$ by fixing unrelated variables, $Loss_1$ in (12) can be simplified as

$$\min_{\mathbf{M}} \sum_{m=0}^o \alpha_m \|\mathbf{W}_m \hat{\mathbf{X}}_m - \mathbf{M} \odot \mathbf{F}\|_F^2 \quad (15)$$

Due to $\mathbf{E} = \mathbf{M} \odot \mathbf{F}$, the solution of $\mathbf{E}$ can be obtained as $\mathbf{E} = \sum_{m=0}^o \alpha_m \mathbf{W}_m \hat{\mathbf{X}}_m$. $\mathbf{F} \in \{-0.5, 1\}$ constrains $\mathbf{M}$ to have same structure with it, and $\mathbf{F} \odot \mathbf{F} \geq \mathbf{F}$, so that $\mathbf{E} \odot \mathbf{F} = (\sum_{m=0}^o \alpha_m \mathbf{W}_m \hat{\mathbf{X}}_m) \odot \mathbf{F} \odot \mathbf{F} \geq \mathbf{M} \odot \mathbf{F}$. Therefore, $\mathbf{M}$ can be addressed as follow:

$$\mathbf{M} = (\sum_{m=0}^o \alpha_m \mathbf{W}_m^T \hat{\mathbf{X}}_m) \odot \mathbf{F} \quad (16)$$

Step 3: Update $\mathbf{V}$ by fixing unrelated variables, $Loss_2$ in (12) is rewritten as

$$\begin{aligned} \min_{\mathbf{V}} \|\mathbf{r}\mathbf{E}^T \mathbf{E} - \mathbf{B}^T \mathbf{V}\|_F^2 \\ \text{s.t. } \mathbf{V}\mathbf{V}^T = n\mathbf{I}_r, \mathbf{V}\mathbf{1} = \mathbf{0}_r \end{aligned} \quad (17)$$

Equation (17) can be further simplified as the following trace form:

$$\max_{\mathbf{V}} Tr\{(\mathbf{r}\mathbf{E}^T \mathbf{E} \mathbf{V}^T) \mathbf{V}\} \quad (18)$$

To address (18), we set a centered matrix $\mathbf{J} = \mathbf{I} - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T$ and $\mathbf{K} = \mathbf{r}\mathbf{E}^T \mathbf{E} \mathbf{B}^T$. The SVD of $\mathbf{K}^T \mathbf{J} \mathbf{K}$ can be performed as:

$$\mathbf{K}^T \mathbf{J} \mathbf{K} = \begin{bmatrix} \mathbf{N} & \bar{\mathbf{N}} \end{bmatrix} \begin{bmatrix} \mathbf{\Psi} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{N} & \bar{\mathbf{N}} \end{bmatrix}^T \quad (19)$$

where $\mathbf{\Psi} \in \mathbb{R}^{\tilde{r} \times \tilde{r}}$ is a diagonal matrix, $\mathbf{N} \in \mathbb{R}^{r \times \tilde{r}}$ and $\bar{\mathbf{N}} \in \mathbb{R}^{r \times (r-\tilde{r})}$ are the corresponding eigenvectors for positive and zero eigenvalues, respectively. $\tilde{r} \leq r$ is rank of $\mathbf{K}^T \mathbf{J} \mathbf{K}$. The solution of $\mathbf{V}$ is:

$$\mathbf{V} = \sqrt{n} \begin{bmatrix} \mathbf{N} & \hat{\mathbf{N}} \end{bmatrix} \begin{bmatrix} \mathbf{R} & \hat{\mathbf{R}} \end{bmatrix}^T \quad (20)$$

where $\hat{\mathbf{N}} \in \mathbb{R}^{r \times (r-r')}$ is an orthogonal matrix from $\bar{\mathbf{N}}$, $\mathbf{R} = \mathbf{J} \mathbf{K}^T \mathbf{N} \mathbf{\Psi}$, and $\hat{\mathbf{R}} \in \mathbb{R}^{n \times (r-r')}$ is an orthogonal random matrix. If the rank of $\mathbf{K}^T \mathbf{J} \mathbf{K}$ is full ($r' = r$), the matrices $\bar{\mathbf{N}}$, $\hat{\mathbf{N}}$ and $\hat{\mathbf{R}}$ are empty.

Step 4: Update $\mathbf{R}$ while fixing other unrelated variables in $Loss_2$ of (12) which can be simplified as

$$\min_{\mathbf{R}} \|\mathbf{B} - \mathbf{R}\mathbf{U}\|_F^2 = \max_{\mathbf{R}} Tr(\mathbf{R}\mathbf{U}\mathbf{B}^T) \quad (21)$$

Let $[\mathbf{Q}_1, \mathbf{\Xi}_1, \mathbf{T}_1] = SVD(\mathbf{R}\mathbf{U}\mathbf{B}^T)$, the solution of $\mathbf{R}$ can be obtained as

$$\mathbf{R} = \mathbf{Q}_1 \mathbf{T}_1^T \quad (22)$$

Step 5: By fixing other parameters, the initial hash codes $\mathbf{B}$ in $Loss_2$ of (12) can be simplified as

$$\min_{\mathbf{B}} \|\mathbf{r}\mathbf{E}^T \mathbf{E} - \mathbf{B}^T \mathbf{V}\|_F^2 + \mu \|\mathbf{B} - \mathbf{R}\mathbf{U}\|_F^2 \quad (23)$$

Due to $\mathbf{V}\mathbf{V}^T = n\mathbf{I}$, $\mathbf{B}$ can be easily obtained as

$$\mathbf{B} = \text{sgn} \left(\frac{1}{n + \mu} (\mathbf{r}\mathbf{V}\mathbf{E}^T \mathbf{E} + \mu \mathbf{R}\mathbf{U}) \right) \quad (24)$$

Step 6: Update $\mathbf{Q}$ in $Loss_3$ is similar to $\mathbf{W}_m$ and $\mathbf{R}$, we let $[\mathbf{C}, \mathbf{S}, \mathbf{E}] = SVD(\mathbf{D}\mathbf{Y}^T \mathbf{B}^T)$, and then $\mathbf{Q}$ can be addressed as:

$$\mathbf{Q} = \mathbf{U}\mathbf{V}^T \quad (25)$$

Step 7: Update diagonal matrix $\mathbf{D}$ by fixing unrelated variables, due to $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}$, $Loss_3$ in (12) can be simplified as

$$\min_{\mathbf{D}} \|\mathbf{Q}\mathbf{V} - \mathbf{D}\mathbf{Y}^T\|_F^2 \quad (26)$$

The element d_i of diagonal matrix $\mathbf{D}$ represents the similarity between i -th row ($\mathbf{v}^{(i)}$) of $\mathbf{Q}\mathbf{B}$ and i -th row ($\mathbf{y}^{(i)}$) of $\mathbf{Y}^T$. Therefore, the result of $d^{(i)}$ can be obtained as $d^{(i)} = \frac{\mathbf{b}^{(i)} \star \mathbf{y}^{(i)}}{\|\mathbf{b}^{(i)}\|_2 \|\mathbf{y}^{(i)}\|_2}$. The solution of $\mathbf{D}$ is represented as

$$d^{(i)} = \begin{cases} 1, & d^{(i)} \geq \text{mean} \\ -1, & d^{(i)} < \text{mean} \end{cases} \quad (27)$$

where mean is the mean value of the $\mathbf{D}$.

Step 8: Since $\mathbf{H}_2$ is the nonlinear projection, it does not have a closed-form solution. The gradient descent is utilized to obtain the iterative solution of $\mathbf{H}_2$ in $Loss_4$ as follows:

$$\mathbf{H}_2^{(t+1)} = \mathbf{H}_2^{(t)} + \gamma \frac{\partial f^{(t)}}{\partial \mathbf{H}_2^{(t)}} \quad (28)$$

where γ is the learning rate, and $\frac{\partial f^{(t)}}{\partial \mathbf{H}_2^{(t)}}$ is the derivative of (12)

w.r.t. $\mathbf{H}_2^{(t)}$ as

$$\begin{aligned} \frac{\partial f^{(t)}}{\partial \mathbf{H}_2^{(t)}} = & -\mathbf{X}_2 (\mathbf{V} - \tanh(\mathbf{H}_2 \mathbf{X}_2)) \\ & \odot (1 - \tanh(\mathbf{H}_2 \mathbf{X}_2) \odot \tanh(\mathbf{H}_2 \mathbf{X}_2)) \end{aligned} \quad (29)$$

Step 9: According to (10), after obtaining $\mathbf{Z}_1 = \text{sgn}(\mathbf{Q}^T \mathbf{D}\mathbf{Y}^T)$ and $\mathbf{Z}_2 = \text{sgn}(\mathbf{H}_2 \mathbf{X}_2^T)$, updating $\mathbf{A}$ in $Loss_5$ of (12) can be simply presented as

$$a^{(i)} = \begin{cases} 1, & \text{sum}(\mathbf{z}_1^{(i)} \odot \mathbf{v}^{(i)}) \geq \text{sum}(\mathbf{z}_2^{(i)} \odot \mathbf{v}^{(i)}) \\ 0, & \text{sum}(\mathbf{z}_1^{(i)} \odot \mathbf{v}^{(i)}) < \text{sum}(\mathbf{z}_2^{(i)} \odot \mathbf{v}^{(i)}) \end{cases} \quad (30)$$

Step 10: Update $\mathbf{B}$ by fixing other variables, the solution of final hash codes $\mathbf{B}$ in $Loss_6$ can be easily obtained as:

$$\mathbf{B} = \text{sgn}((\mathbf{Z}\mathbf{Z}^T + \mu\mathbf{I})^{-1}(\mathbf{f}\mathbf{Z}\mathbf{E}^T \mathbf{E} + \mu\mathbf{R}\mathbf{U})) \quad (31)$$

H. Final Hash Functions Learning

To learn the final hash functions for the out-of-sample extension from the final hash codes, literature typically employs kernel logistic regression or ridge regression [16], [29], [45]. To reduce computational complexity and mitigate the issue of over-fitting of hash functions, the ridge regression is introduced to learn the final hash functions $\mathbf{P}_m \in \mathbb{R}^{r \times d_m}$, given as:

$$\min_{\mathbf{P}_m} \|\mathbf{B} - \mathbf{P}_m \mathbf{X}_m\|_F^2 + \theta \|\mathbf{P}_m\|_F^2 \quad (32)$$

where $\theta > 0$ is the penalty parameter. We set the derivative of $\mathbf{P}_m$ in (32) to zero, obtaining the solution as:

$$\mathbf{P}_m = \mathbf{B}\mathbf{X}_m^T (\mathbf{X}_m \mathbf{X}_m^T + \theta \mathbf{I})^{-1} \quad (33)$$

Algorithm 1: Optimization process of RCMH.

Input: The training data $\mathbf{X}_m$, label matrix $\mathbf{L}$, existing hash codes $\mathbf{U}$, bit length of final hash codes r , parameters α_0, μ, θ and learning rate γ .

Initialization: Initializing hash codes $\mathbf{B}$ by clipping or padding the existing hash codes $\mathbf{U}$, normalizing the training data $\mathbf{X}_m$ into $\hat{\mathbf{X}}_m$

procedure

1. Label Augmentation

Step 1. Update orthogonal projection $\mathbf{W}_m$ by using (14);

Step 2. Update label weight matrix $\mathbf{M}$ by using (16);

2. Initial Final Hash Codes

Step 3. Update real-valued matrix $\mathbf{V}$ with (20);

Step 4. Update orthogonal matrix $\mathbf{R}$ with (22);

Step 5. Initial final hash codes $\mathbf{B}$ by using (24);

3. Linear-Nonlinear Competitive Reconstruction

Step 6. Update orthogonal matrix $\mathbf{Q}$ by using (25);

Step 7. Update diagonal matrix $\mathbf{D}$ by using (27);

Step 8. Update nonlinear projection $\mathbf{H}_2$ by using (29);

Step 9. Update competitive matrix $\mathbf{A}$ by using (30);

4. Regenerated Final Hash Codes

Step 10. Update final hash codes $\mathbf{B}$ by using (31);

end procedure

Return: The hash codes matrix $\mathbf{B}$, obtained final hash functions $\mathbf{P}_m$ with (33)

Out-of-Sample Extension: Once we obtain the final hash function $\mathbf{P}_m$, the hash codes for new arrived query data $\mathbf{X}_m^{query}$ in m -th modality can be represented as follows:

$$\mathbf{B}_m^{query} = \text{sgn}(\mathbf{P}_m \mathbf{X}_m^{query}) \quad (34)$$

The results of comparing $\mathbf{B}_m^{query}$ with the training hash codes $\mathbf{B}$ are sorted in ascending order under the Hamming distance to obtain final retrieval results.

I. Computational Complexity

To demonstrate the high efficiency of RCMH, we conduct a detailed computational complexity analysis in this subsection. The analysis of all variables is divided into four stages as shown in Algorithm 1. In the label augmentation stage, the computational complexity of $\mathbf{W}_m$ in (14) is $O(d_m cn + d_m^2 n + cn)$ and $\mathbf{M}$ in (16) is $O((ocd_m n + cn))$. During the initial hash codes stage, the computational complexity for solving $\mathbf{V}$ in (20) is $O(r^3 + r^2 n + rcn + rn)$, addressing $\mathbf{R}$ in (22) is $O(rfn + r^2 n + r^2 f)$ and solving $\mathbf{B}$ in (24) is $O(frn + rcn + rn)$. In the linear-nonlinear competitive reconstruction stage, the computational complexity of addressing $\mathbf{Q}$ in (25) is $O(d_1^2 n + d_1 rn)$, addressing $\mathbf{D}$ in (27) is $O(d_1 n)$, and solving $\mathbf{H}_2$ in (29) is $O(rn + d_2 rn)$. In the regenerated final hash codes stage, the computational complexity of addressing $\mathbf{B}$ in (31) is $O(r^3 + r^2 n + rcn + rn + frn)$. It is worth noting that the computational complexity of RCMH scales linearly with n because $r, f, c, d_m \ll n$. This demonstrates the high efficiency of our proposed method.

IV. EXPERIMENTS

A. Datasets

To validate the performance of RCMH, we conducted extensive experiments on four widely adopted datasets, i.e., IAPR TC-12,¹ MIR-Flickr,² NUS-WIDE³ and MS-COCO.⁴ Until now, these are the publicly available large-scale multi-label cross-modal datasets that are fully paired and labeled.

IAPR TC-12 [69] includes 20,000 image-text pairs labeled by 255 semantic categories. Each image is represented as a 512-dimensional GIST feature vector, and each text is represented as a 2,912-dimensional bag-of-words vector. A set of 2,000 pairs was randomly selected for testing, and the rest was used for training.

MIR-Flickr [70] contains 25,000 image-text pairs from Flickr and is labeled by 24 semantic categories. After removing pairs with fewer than 20 tags in text, it includes 20,015 pairs, of which 2,000 were chosen as a test set and 18,015 as a training set. Each image is represented as a 512-dimensional GIST feature vector, and each text is represented as a 1,386-dimensional bag-of-words vector.

NUS-WIDE [71] is a dataset of 269,648 image-text pairs labeled with 81 semantic categories, obtained from Flickr. For the experiments, only the 10 most common labels and their corresponding 186,577 pairs were selected, with 2,000 instances of them used for testing and the rest for training. Each image is represented as a 500-dimensional SIFT feature vector, and each text is represented as a 1,000-dimensional binary tagging vector.

MS-COCO [72] is a dataset that collects more than 300,000 image-text pairs from real-world scenarios. Each pair is labeled with one or more of the 80 semantic concepts. By removing images without texts or labels on MS-COCO 2014, 82,081 image-text pairs can be obtained; 2,000 pairs were randomly selected as the test set, and the remaining pairs were considered as the training set. According to the paper [73], the image is first represented as 4,096-dimensional features extracted using the VGG network implemented in Caffe and then is reduced to a 512-dimensional feature by PCA. Similarly, the text is first represented as a 20,629-dimensional bag-of-words vector by the natural language toolkit (NLTK) and then is also reduced to a 512-dimensional feature by PCA.

B. Evaluation Metrics

To comprehensively evaluate the performance of RCMH, we conducted experiments on typical cross-modal retrieval tasks, i.e., image-to-text ($I \rightarrow T$) and text-to-image ($T \rightarrow I$) tasks, which are commonly used in the literature [42], [43], [74]. To assess the performance of the cross-modal retrieval tasks, we utilized two evaluation metrics: mean average precision (MAP) and MAP of top 100 instances (TOP@100) [40], [41], [75], [76]. MAP provides the average precision of all relevant instances, while

¹<https://www.imageclef.org/photodata>

²<http://press.liacs.nl/mirflickr/mirdownload.html>

³<http://lms.comp.nus.edu.sg/research/NUS-WIDE.htm>

⁴<http://mscoco.org/home/>

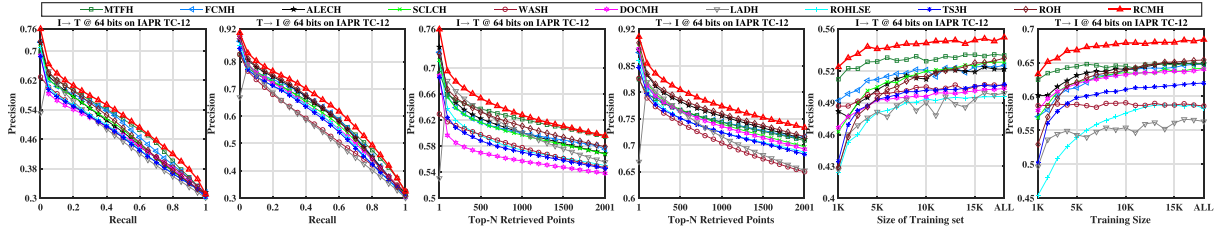


Fig. 5. Curves of Precision-Recall, Top-N, and Training-Size precision of RCMH @ 64 bits on IAPR TC-12.

TOP@100 focuses on the average precision of the top 100 relevant instances. Higher values of MAP and TOP@100 indicate superior retrieval performance of the proposed method. In detail, we also presented comprehensive results in this paper using another three evaluation metrics: Precision-Recall, Top-N Precision, and Training-Size Precision curves. The Precision-Recall curves demonstrate the relationship between precision and recall for all instances, whereas the Top-N Precision curves illustrate the precision of retrieving the N most relevant instances. The Precision-Size curves highlight the variation in precision as the size of the training data increases. Notably, higher values of the curve's height signify improved performance of the proposed method.

C. Baseline Methods and Implementation Details

In this study, we conducted a comparative analysis of RCMH with ten cross-modal hashing baselines. These SOTA methods, listed in chronological order, are MTFH (TPAMI₂₀₂₁) [45], FCMH (TCYB₂₀₂₂) [46], SCLCH (TIP₂₀₂₂) [47], ALECH (TKDE₂₀₂₃) [41], WASH (TKDE₂₀₂₃) [42], DOCMH (PR₂₀₂₄) [43], LADH (ACMTKDD₂₀₂₄) [40], ROHLSE (PR₂₀₂₄) [48], TS3H (TNNLS₂₀₂₅) [44], ROH (TNNLS₂₀₂₅) [49]. It is important to note that the existing hash codes used for RCMH were obtained from FCMH. The source codes for all methods were obtained from their respective websites and implemented using the recommended parameters. To ensure an accurate evaluation of all methods, we conducted each experiment 10 times and averaged the results to obtain the outcomes. For RCMH, we did not iterate over all variables, as it can quickly reach convergence based on the existing hash codes obtained from a convergent method. Thus, its iteration was set to 1. The parameter configuration for RCMH was analyzed and recorded in Section IV-L. All experiments were executed using MATLAB on a system equipped with an AMD Ryzen 9 9950X processor running at 4.3 GHz with 16 cores, an NVIDIA RTX 5090 graphics card, 96 GB RAM, and a 64-bit Windows 10 operating system. Due to space constraints, we moved some experiments to the appendix (available online), such as improvements and robustness on CLIP-based methods with RCMH in Appendix B (available online), comparison of RCMH with shallow baseline under CLIP features in Appendix C (available online), comparisons with large-scale image retrieval in Appendix D (available online), improvements on asymmetric deep hashing techniques in Appendix E (available online), semantic gaps reduction with RCMH in Appendix F (available online) and linear-nonlinear validation in Appendix G, available online.

TABLE II
MAP AND TOP@100 RESULTS OF RCMH AND BASELINES ON IAPR TC-12.
THE BEST PERFORMANCE IS IN BOLDFACE, THE SECOND-BEST PERFORMANCE IS IN UNDERLINE

Tasks	Datasets	IAPR TC-12							
		MAP				TOP@100			
		16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits
I → T	MTFH	0.4918	0.5126	0.5347	0.5528	0.6146	0.6634	0.6844	0.7056
	FCMH	0.4929	0.5087	0.5309	0.5402	0.6205	0.6409	0.6527	0.6596
	SCLCH	0.4779	0.5086	0.5287	0.5390	0.5382	0.6323	0.6676	0.6844
	ALECH	0.4777	0.5043	0.5204	0.5329	0.5792	0.6441	0.6769	0.6956
	WASH	0.4827	0.4985	0.5080	0.5137	0.5654	0.5968	0.6184	0.6315
	DOCMH	0.4618	0.4832	0.4950	0.5052	0.5923	0.6202	0.6114	0.6322
	LADH	0.4718	0.4827	0.4982	0.5112	0.6684	0.6754	<u>0.6912</u>	0.6898
	ROHLSE	0.4697	0.4914	0.5057	0.5173	0.5598	0.6352	0.6582	0.6497
	TS3H	0.4657	0.4888	0.5084	0.5162	0.5554	0.6052	0.6368	0.6502
	ROH	0.4811	0.5074	0.5317	0.5465	0.5903	0.6362	0.6813	0.7041
	RCMH	0.5113	0.5348	0.5506	0.5640	<u>0.6483</u>	0.6841	0.7026	0.7248
T → I	MTFH	0.5792	0.6271	0.6493	0.6637	0.7011	0.7775	0.8098	0.8335
	FCMH	<u>0.5855</u>	0.6132	0.6496	0.6701	0.7676	0.7976	0.8183	0.8292
	SCLCH	0.5693	0.6066	0.6408	0.6564	0.7279	0.7972	0.8394	0.8570
	ALECH	0.5752	0.6163	0.6472	0.6672	0.7515	<u>0.8193</u>	<u>0.8495</u>	0.8660
	WASH	0.5454	0.5705	0.5863	0.5953	0.7168	0.7674	0.8020	0.8196
	DOCMH	0.5650	0.6010	0.6288	0.6535	0.7736	0.8140	0.8422	0.8594
	LADH	0.5301	0.5512	0.5716	0.5936	<u>0.7892</u>	0.8099	0.8251	0.8377
	ROHLSE	0.5576	0.5964	0.6174	0.6312	0.7275	0.7849	0.8043	0.8123
	TS3H	0.5494	0.5905	0.6022	0.6321	0.7231	0.7798	0.8104	0.8231
	ROH	0.5758	0.6133	<u>0.6547</u>	<u>0.6789</u>	0.7575	0.8133	0.8488	<u>0.8701</u>
	RCMH	0.6064	0.6514	0.6852	0.7083	0.7986	0.8422	0.8697	0.8851

D. Results and Discussions

IAPR TC-12: Table II and Fig. 5 present the evaluation results of all methods on IAPR TC-12. It is clear that RCMH outperforms baselines on all evaluation metrics and demonstrates significant advancements, showcasing improvements in MAP and TOP@100 by approximately 2% over the baselines. The main reason may be that RCMH can explore more similarity information from modality features into the augmented label information, which in turn improves the quality of hash codes in cross-modal retrieval tasks. The only exception is the 16-bit TOP@100 results for I → T tasks, which may be due to LADH's use of the $L_{2,1}$ norm for generating better hash codes, giving it an advantage in low-bit tasks. The results in Fig. 5 also show that the Precision-Recall and Top-N curves align with the MAP and TOP@100 results in Table II. As the training size increases, RCMH consistently shows superior MAP results compared to the baselines, indicating its strong performance on both small and large-scale datasets.

MIR-Flickr: The results for all methods on MIR-Flickr are summarized in Table III and Fig. 6. RCMH consistently outperforms all baselines, with improvements of 1 – 2% in MAP metrics. This superior performance can be attributed to the linear-nonlinear competitive reconstruction, which effectively captures

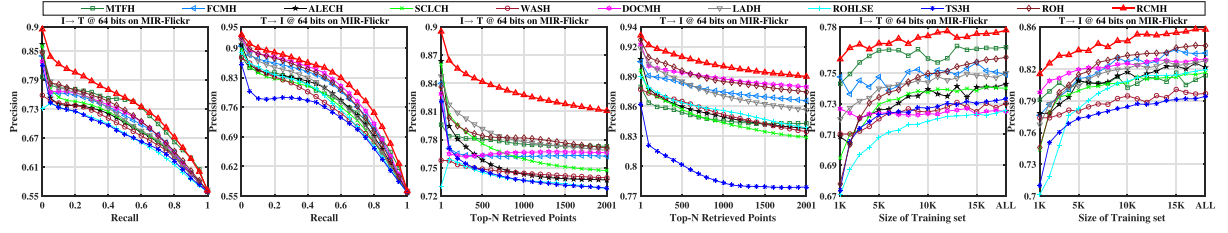


Fig. 6. Curves of Precision-Recall, Top-N, and Training-Size precision of RCMH @ 64 bits on MIR-Flickr.

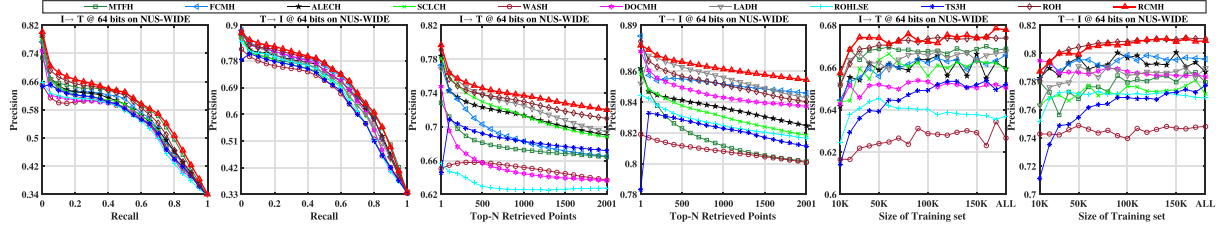


Fig. 7. Curves of Precision-Recall, Top-N, and Training-Size precision of RCMH @ 64 bits on NUS-WIDE.

TABLE III

MAP AND TOP@100 RESULTS OF RCMH AND BASELINES ON MIR-Flickr. THE BEST PERFORMANCE IS IN BOLDFACE, THE SECOND-BEST PERFORMANCE IS IN UNDERLINE

Tasks	Datasets	MIR-Flickr							
		MAP				TOP@100			
	Methods	16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits
I → T	MTFH	0.7488	0.7606	0.7667	0.7688	0.7712	0.7913	0.8004	0.7982
	FCMH	0.7405	0.7475	0.7517	0.7544	0.7833	0.7844	0.7866	0.7909
	SCLCH	0.7322	0.7393	0.7407	0.7477	0.7818	0.7896	0.8012	0.8228
	ALECH	0.7317	0.7356	0.7393	0.7406	0.7949	0.8048	0.8122	0.8142
	WASH	0.7231	0.7269	0.7303	0.7312	0.7411	0.7459	0.7508	0.7534
	DOCMH	0.7287	0.7407	0.7385	0.7419	0.7804	0.7511	0.7502	0.7456
	LADH	0.7339	0.7382	0.7418	0.7438	0.8039	<u>0.8051</u>	<u>0.8175</u>	<u>0.8236</u>
	ROHLS	0.7241	0.7298	0.7349	0.7358	0.7679	0.7878	0.8067	0.8101
	TS3H	0.7277	0.7340	0.7411	0.7405	0.7800	0.7950	0.8072	0.8181
	ROH	0.7338	0.7502	0.7541	0.7576	0.7845	0.7934	0.8047	0.8198
	RCMH	0.7568	0.7691	0.7755	0.7848	0.8278	0.8519	0.8617	0.8774
T → I	MTFH	0.7929	0.8040	0.8141	0.8116	0.8483	0.8602	0.8795	0.8721
	FCMH	0.8217	0.8318	0.8367	0.8401	0.8897	0.8932	0.8951	0.8952
	SCLCH	0.8002	0.8100	0.8138	0.8157	0.8628	0.8754	0.8730	0.8813
	ALECH	0.8063	0.8157	0.8203	0.8234	0.8768	0.8800	0.8846	0.8850
	WASH	0.7876	0.7960	0.7975	0.8008	0.8604	0.8682	0.8730	0.8737
	DOCMH	0.8214	0.8293	0.8365	0.8391	0.8952	0.9003	0.8965	0.8971
	LADH	0.8110	0.8192	0.8246	0.8264	0.8947	0.8980	0.9032	0.9046
	ROHLS	0.8083	0.8152	0.8181	0.8196	0.8844	0.8920	0.8974	0.9004
	TS3H	0.7999	0.8082	0.8122	0.8138	0.8579	0.8677	0.8752	0.8757
	ROH	0.8164	0.8339	0.8421	0.8445	0.8955	0.9054	0.9081	0.9120
	RCMH	0.8322	0.8488	0.8578	0.8666	0.9010	0.9144	0.9184	0.9233

TABLE IV

MAP AND TOP@100 RESULTS OF RCMH AND BASELINES ON NUS-WIDE. THE BEST PERFORMANCE IS IN BOLDFACE, THE SECOND-BEST PERFORMANCE IS IN UNDERLINE

Tasks	Datasets	NUS-WIDE							
		MAP				TOP@100			
	Methods	16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits
I → T	MTFH	0.6380	0.6615	0.6695	0.6724	0.6916	0.7224	0.7407	0.7457
	FCMH	0.6458	0.6572	0.6626	0.6671	0.7322	0.7544	0.7495	0.7547
	SCLCH	0.6426	0.6539	0.6640	0.6684	0.7425	0.7531	0.7681	0.7699
	ALECH	0.6389	0.6568	0.6601	0.6650	0.7354	0.7533	0.7601	0.7650
	WASH	0.6251	0.6270	0.6283	0.6306	0.6325	0.6525	0.6625	0.6731
	DOCMH	0.6424	0.6480	0.6507	0.6538	0.7182	0.7346	0.7218	0.7551
	LADH	0.6448	0.6539	0.6609	0.6678	<u>0.7462</u>	0.7632	<u>0.7760</u>	<u>0.7838</u>
	ROHLS	0.6335	0.6419	0.6410	0.6489	0.7237	0.7511	0.7540	0.7693
	TS3H	0.6371	0.6432	0.6534	0.6532	0.7093	0.7240	0.7289	0.7272
	ROH	0.6533	0.6678	0.6714	0.6780	0.7401	0.7546	0.7711	0.7792
	RCMH	0.6573	0.6688	0.6781	0.6854	0.7471	<u>0.7560</u>	0.7818	0.7883
T → I	MTFH	0.7427	0.7717	0.7846	0.7843	0.7976	0.8284	0.8541	0.8529
	FCMH	0.7792	0.7903	0.7991	0.8037	0.8552	<u>0.8701</u>	0.8693	0.8736
	SCLCH	0.7570	0.7776	0.7778	0.7880	0.8346	0.8485	0.8440	0.8545
	ALECH	0.7705	0.7871	0.7932	0.7964	0.8522	0.8579	0.8604	0.8606
	WASH	0.7340	0.7421	0.7466	0.7479	0.7973	0.8125	0.8209	0.8237
	DOCMH	0.7716	0.7795	0.7835	0.7885	0.8447	0.8573	0.8472	0.8720
	LADH	0.7724	0.7845	0.7913	0.7896	0.8508	0.8630	0.8680	0.8715
	ROHLS	0.7624	0.7750	0.7758	0.7854	0.8521	0.8653	0.8687	0.8719
	TS3H	0.7557	0.7687	0.7753	0.7802	0.8232	0.8291	0.8295	0.8298
	ROH	0.7801	0.8041	0.8113	0.8091	0.8516	0.8582	0.8670	0.8680
	RCMH	0.7836	<u>0.8013</u>	<u>0.8095</u>	0.8218	0.8612	0.8732	0.8708	0.8758

both linear and nonlinear features from high-dimensional data into the final hash codes. This makes RCMH well-suited for handling different tasks and achieving better MAP results. Notably, RCMH consistently achieves optimal results even with a small training set size of 1 K, suggesting that it can maintain stable performance as the number of instances increases on MIR-Flickr.

NUS-WIDE: The results on NUS-WIDE are presented in Table IV and Fig. 7. RCMH achieves the best MAP results, but exhibits lower TOP@100 results compared to the baselines. LADH stands out in terms of TOP@100 results, similar to the observation on IAPR TC-12. ROH, leveraging modality-specific features with online learning, captures more local similarity from low-dimensional SIFT image features and thus obtains better

T → I MAP results in 32 bits and 64 bits. However, the Top-N curves on NUS-WIDE show that RCMH still outperforms baselines over the top 100 points, demonstrating its high effectiveness in retrieving the top 2001 instances. In addition, RCMH gets better MAP results with the increase of training sample size, which is similar to online learning methods (e.g., ROH and DOCMH). This indicates that RCMH is capable of handling datasets with more training samples.

MS-COCO: The results for all methods on MS-COCO are presented in Table V and Fig. 8. RCMH demonstrates superior performance over the baselines across all evaluation metrics on this large-scale dataset. The Precision-Size curves in Fig. 8 further highlight RCMH's capability to retrieve instances with higher accuracy across diverse modalities. Compared to

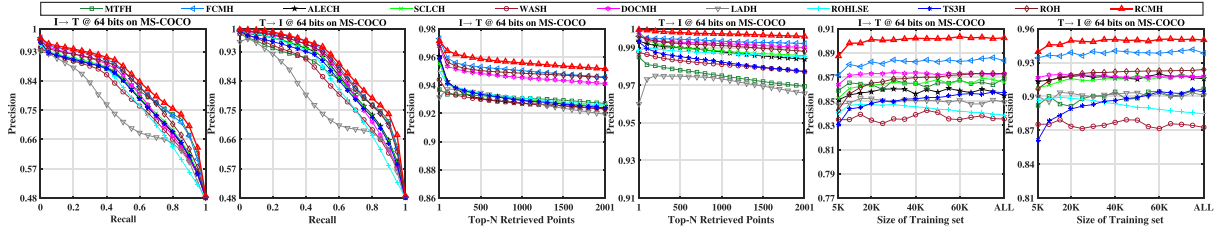


Fig. 8. Curves of Precision-Recall, Top-N, and Training-Size precision of RCMH @ 64 bits on MS-COCO.

TABLE V

MAP AND TOP@100 RESULTS OF RCMH AND BASELINES ON MS-COCO. THE BEST PERFORMANCE IS IN BOLDFACE, THE SECOND-BEST PERFORMANCE IS IN UNDERLINE

Tasks	Datasets	MS-COCO							
		MAP				TOP@100			
		16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits
I → T	Methods	16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits
	MTFH	0.8351	0.8596	0.8667	0.8709	0.9101	0.9262	0.9391	0.9380
	FCMH	<u>0.8502</u>	<u>0.8735</u>	<u>0.8868</u>	<u>0.8964</u>	0.9406	0.9489	0.9505	0.9579
	SCLCH	0.8414	0.8588	0.8658	0.8708	0.9413	0.9448	0.9498	0.9507
	ALECH	0.8341	0.8498	0.8595	0.8638	0.9405	0.9440	0.9471	0.9440
	WASH	0.8166	0.8329	0.8371	0.8383	0.9159	0.9309	0.9349	0.9378
	DOCMH	0.8300	0.8529	0.8649	0.8699	<u>0.9428</u>	<u>0.9519</u>	<u>0.9566</u>	<u>0.9598</u>
	LADH	0.7747	0.7979	0.8063	0.8150	0.8410	0.8969	0.9295	0.9282
	ROHLSE	0.8273	0.8419	0.8560	0.8603	0.9364	0.9451	0.9501	0.9524
	TS3H	0.8312	0.8488	0.8569	0.8614	0.9328	0.9424	0.9462	0.9464
	ROH	0.8355	0.8598	0.8717	0.8749	0.9424	0.9439	0.9480	0.9475
	RCMH	0.8679	0.8886	0.9031	0.9134	0.9521	0.9613	0.9641	0.9685
T → I	MTFH	0.8558	0.8853	0.9023	0.9083	0.9348	0.9724	0.9832	0.9867
	FCMH	<u>0.8952</u>	<u>0.9237</u>	<u>0.9388</u>	<u>0.9428</u>	<u>0.9913</u>	<u>0.9921</u>	<u>0.9944</u>	<u>0.9954</u>
	SCLCH	0.8884	0.9068	0.9147	0.9208	0.9899	0.9917	0.9941	0.9940
	ALECH	0.8824	0.9040	0.9176	0.9258	0.9889	0.9926	0.9932	0.9934
	WASH	0.8505	0.8689	0.8741	0.8756	0.9722	0.9845	0.9877	0.9887
	DOCMH	0.8758	0.9015	0.9143	0.9187	0.9874	0.9924	0.9939	0.9950
	LADH	0.8251	0.8485	0.8543	0.8648	0.8651	0.9330	0.9723	0.9764
	ROHLSE	0.8769	0.8907	0.9077	0.9115	0.9912	0.9921	0.9938	0.9948
	TS3H	0.8769	0.8950	0.9045	0.9105	0.9842	0.9882	0.9907	0.9922
	ROH	0.8788	0.9098	0.9228	0.9234	0.9897	<u>0.9936</u>	<u>0.9947</u>	0.9943
	RCMH	0.9086	0.9341	0.9531	0.9637	0.9957	0.9982	0.9986	0.9995

traditional hashing methods, RCMH significantly reduces the semantic gap between modalities by enhancing existing hash codes, thereby performing better in cross-modal retrieval. This is further verified in Section IV-E and Appendix F, available online. It is also interesting to note that all methods achieve better results on MS-COCO than on other datasets. This can be attributed to the use of image features extracted by the deep convolutional network VGG, which inherently contains richer nonlinear and local information compared to linear models like GIST and SIFT. As a result, the shallow methods employed in this study show proficiency in handling retrieval tasks with features extracted from deep neural networks. Furthermore, there is potential for RCMH to be transformed into deep end-to-end models to more effectively address cross-modal retrieval tasks, as demonstrated in Section IV-M.

E. Gain Improvements to Existing SOTAs

As previously demonstrated, RCMH can effectively enhance existing methods using their hash codes. To evaluate the scalability and efficacy of RCMH, we extended it to nine SOTA baselines: FCMH, SCLCH, ALECH, WASH, DOCMH, LADH, ROHLSE, TS3H, and ROH. MTFH was excluded since it uses kernel logistic regression to learn hash functions, which is unsuitable for RCMH. Specifically, we discarded the kernelizations

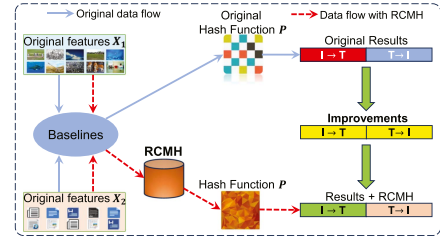


Fig. 9. Illustration of improvements in baselines caused by RCMH.

and used the original features X_m as input for the selected baselines. We then adjusted the over-fitting parameters to obtain their best results, labeled as ‘Original Results’ in Fig. 9. After obtaining their hash codes, we applied RCMH to enhance these codes, achieving retrieval results labeled as ‘Results + RCMH’ in Fig. 9. The difference between ‘Original Results’ and ‘Results + RCMH’, termed ‘Improvement’, is shown as yellow bars in Fig. 9. Further results for ‘Results + RCMH’ are presented in Fig. 10, where red bars indicate original I → T results, blue bars show T → I results, and yellow bars represent the improvements between them.

From Fig. 10, it is evident that all baselines achieve improvements when RCMH is used to regenerate their hash codes. Methods such as ALECH, WASH, DOCMH, LADH, and ROHLSE see approximately 5% improvements when enhanced with RCMH, highlighting RCMH’s significant effectiveness in boosting cross-modal retrieval performance. This can largely be attributed to RCMH’s reduction of the semantic gap between different modalities, as visualized in Fig. A-2 of Appendix F, available online. However, the marginal utility of RCMH tends to decrease; the better the original baselines, the fewer improvements RCMH can provide. This is because these baselines’ original results are already near the shallow features’ limit (e.g., PCA, SIFT, and GIST). Substantial further improvements may require deep features like CNN features, as discussed in Section IV-M. On the other hand, RCMH can achieve good retrieval results even if the baselines have poor performance (e.g., ALECH and WASH), which shows the robustness of RCMH. In summary, these results indicate that RCMH can be easily integrated with other SOTA methods to enhance their retrieval performance with high efficiency, which can also be verified in the Section IV-G.

F. Bit Length Conversion and Robustness of RCMH

To demonstrate RCMH’s effectiveness in bit length conversion, we presented retrieval results for converting existing hash codes to final hash codes of different bit lengths, such as from 16-bit to 128-bit and vice versa. In this subsection, the bit length

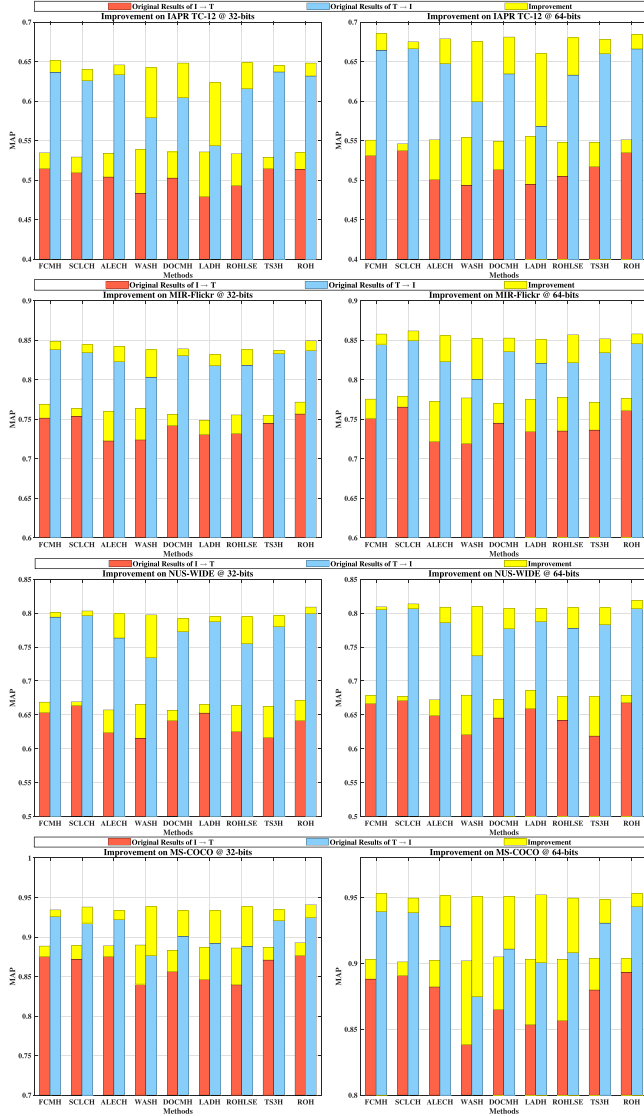


Fig. 10. The improvements of the original baselines' MAP compared with results of baselines by RCMH.

conversion process is recorded by first fixing the bit length of the existing hash codes and converting them into final hash codes of another bit length, then changing the bit length of the existing hash codes. Additionally, we replaced the existing hash codes from FCMH with 'Random', 'Normalize', and 'Zeros' initialization methods to test RCMH's robustness in retrieval tasks.

As shown in Table VI, the results are largely unrelated to the bit length r of existing hash codes but depend on the bit length f of the final hash codes. Converting between the same bit lengths does not yield significantly better results than upgrading from low-dimensional to high-dimensional or reducing from high-dimensional to low-dimensional hash codes. This indicates that RCMH can convert existing hash codes into arbitrary bit lengths with stable performance. 'Random', 'Normalize', and 'Zeros' methods achieve worse results than the standard bit length conversion, revealing that low-quality existing hash codes may negatively impact the results. This observation is further verified

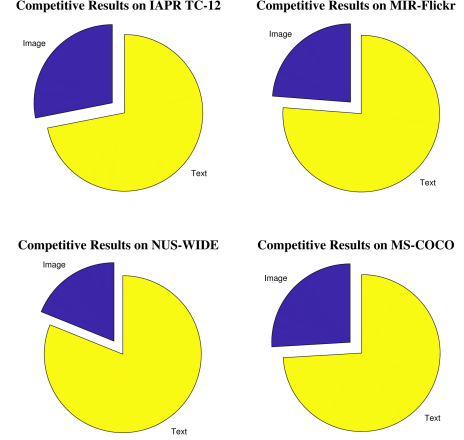


Fig. 11. The competitive results of image and text features in RCMH.

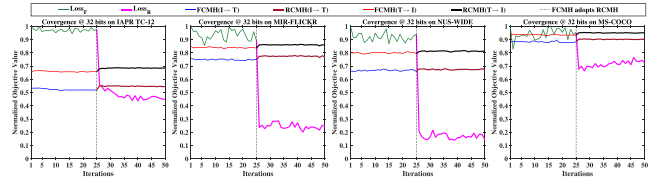


Fig. 12. The convergence and MAP results of FCMH and RCMH.

in Section IV-E. However, even with poor-quality existing hash codes as input, RCMH can still achieve satisfactory results, demonstrating its independence and robustness in real-world retrieval tasks.

G. High Efficiency of RCMH

To demonstrate the efficiency of RCMH as a highly scalable plugin, we compared its time and memory costs with those of the baselines. All experimental results were averaged over 10 runs. Notably, RCMH did not retrain FCMH but merely utilized the hash codes generated by FCMH. Table VII lists the training time (seconds) and memory cost (MB) required by each method to obtain hash codes. For MTFH, the training size was set to $n = 50$ K on the NUS-WIDE and MS-COCO datasets due to its computational complexity of $O(n^2)$.

As shown in Table VII, RCMH achieves the shortest training time and minimum memory usage. This efficiency is attributed to RCMH's ability to obtain results without iterating over all variables, thereby enhancing training speed and reducing memory costs through simplified variable solutions. Combined with the MAP results reported in Table II, III, IV, and V, RCMH delivers the best retrieval performance with the least training time and memory usage. This indicates that using RCMH to change the bit length of baselines, instead of retraining their models, is both faster and more effective in improving MAP results. Thus, RCMH can assist more baselines in handling complex real-life retrieval tasks.

H. Linear-Nonlinear Competitive Strategy Analysis

To analyze the influence of linear image features and nonlinear text features, experiments were conducted to compare RCMH's

TABLE VI
THE BIT LENGTH CONVERSION AND ROBUSTNESS OF RCMH WITH DIFFERENT BIT LENGTHS. CORRESPONDING TO THE RESULTS IN TABLE II, III, IV AND V ARE IN BOLDFACE

Task	Bit Length r	IAPR TC-12				MIR-Flickr				NUS-WIDE				MS-COCO			
		16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits
$I \rightarrow T$	16 bits	0.5113	0.5360	0.5528	0.5631	0.7568	0.7674	0.7788	0.7833	0.6573	0.6697	0.6790	0.6853	0.8679	0.8887	0.9047	0.9137
	32 bits	0.5093	0.5348	0.5538	0.5636	0.7541	0.7691	0.7793	0.7841	0.6534	0.6688	0.6817	0.6848	0.8668	0.8886	0.9047	0.9130
	64 bits	0.5146	0.5353	0.5506	0.5637	0.7512	0.7678	0.7755	0.7848	0.6557	0.6704	0.6781	0.6856	0.8687	0.8907	0.9031	0.9132
	128 bits	0.5124	0.5364	0.5524	0.5640	0.7501	0.7678	0.7771	0.7848	0.6553	0.6698	0.6787	0.6854	0.8676	0.8906	0.9045	0.9134
	Random	0.5061	0.5359	0.5483	0.5613	0.7516	0.7642	0.7768	0.7820	0.6516	0.6624	0.6726	0.6807	0.8579	0.8859	0.9001	0.9082
	Normalize	0.5063	0.5309	0.5470	0.5574	0.7482	0.7640	0.7728	0.7790	0.6512	0.6603	0.6709	0.6790	0.8604	0.8843	0.8999	0.9070
$T \rightarrow I$	Zeros	0.5050	0.5305	0.5463	0.5597	0.7487	0.7563	0.7723	0.7780	0.6487	0.6569	0.6687	0.6804	0.8581	0.8707	0.8906	0.9073
	16 bits	0.6064	0.6500	0.6826	0.7048	0.8322	0.8478	0.8610	0.8671	0.7831	0.8005	0.8114	0.8195	0.9086	0.9339	0.9522	0.9636
	32 bits	0.6044	0.6514	0.6823	0.7043	0.8329	0.8488	0.8609	0.8671	0.7792	0.8013	0.8127	0.8187	0.9081	0.9341	0.9533	0.9629
	64 bits	0.6065	0.6507	0.6852	0.7046	0.8330	0.8501	0.8578	0.8666	0.7826	0.8004	0.8095	0.8196	0.9105	0.9365	0.9531	0.9632
	128 bits	0.6055	0.6498	0.6820	0.7083	0.8296	0.8467	0.8599	0.8668	0.7769	0.7980	0.8109	0.8218	0.9097	0.9367	0.9529	0.9637
	Random	0.6006	0.6477	0.6806	0.7017	0.8291	0.8463	0.8591	0.8646	0.7760	0.7934	0.8057	0.8124	0.8999	0.9319	0.9483	0.9582
	Normalize	0.5991	0.6430	0.6763	0.6987	0.8275	0.8424	0.8555	0.8620	0.7783	0.7904	0.8041	0.8129	0.9019	0.9301	0.9471	0.9566
	Zeros	0.5950	0.6438	0.6745	0.6996	0.8268	0.8368	0.8534	0.8596	0.7759	0.7871	0.8015	0.8099	0.9001	0.9170	0.9414	0.9566

TABLE VII
THE TRAINING TIME AND MEMORY COST COMPARISONS OF RCMH WITH BASELINES. THE BEST PERFORMANCES ARE IN BOLDFACE, THE SECOND-BEST PERFORMANCE IS IN UNDERLINE

Task	Methods	IAPR TC-12		MIR-Flickr		NUS-WIDE		MS-COCO	
		32 bits	64 bits	32 bits	64 bits	32 bits	64 bits	32 bits	64 bits
Training Times (Seconds) and Memory Cost (MB)	MTFH	169.42s (10504MB)	349.68s (10549MB)	173.19s (9392MB)	350.99s (9408MB)	448.20s (39400MB)	873.12s (39451MB)	423.68s (38488MB)	855.62s (38638MB)
	FCMH	4.86s (5524MB)	5.49s (5533MB)	1.84s (5443MB)	1.97s (5444MB)	13.26s (13029MB)	16.24s (13094MB)	9.89s (7164MB)	11.3s (7163MB)
	SCLCH	<u>2.77s</u> (5806MB)	<u>3.11s</u> (5830MB)	<u>1.08s</u> (5130MB)	<u>1.50s</u> (5139MB)	<u>7.69s</u> (9868MB)	<u>9.63s</u> (10193MB)	<u>5.18s</u> (6962MB)	<u>6.54s</u> (7092MB)
	ALECH	2.52s (6880MB)	2.79s (6905MB)	1.53s (5774MB)	1.62s (5771MB)	10.38s (15356MB)	12.27s (15508MB)	5.35s (8955MB)	7.05s (8947MB)
	WASH	3.35s (7070MB)	3.67s (7056MB)	1.39s (5920MB)	1.58s (5916MB)	13.28s (14084MB)	14.32s (14084MB)	7.47s (8224MB)	8.61s (8205MB)
	DOCMH	50.96s (5480MB)	52.18s (5463MB)	7.73s (5091MB)	8.20s (5088MB)	39.93s (6949MB)	42.05s (6952MB)	12.47s (5588MB)	13.39s (5598MB)
	LADH	17.54s (9162MB)	18.46s (9165MB)	4.38s (8099MB)	4.58s (8102MB)	47.35s (18756MB)	50.22s (18809MB)	26.39s (11598MB)	27.36s (11605MB)
	ROHLSE	20.04s (9861MB)	21.27s (9862MB)	7.89s (8290MB)	8.94s (8299MB)	19.19s (18940MB)	20.89s (18947MB)	11.26s (11122MB)	12.34s (11126MB)
	TS3H	13.45s (9376MB)	13.91s (9385MB)	3.71s (8207MB)	3.85s (8215MB)	30.13s (18376MB)	31.19s (18465MB)	13.61s (11429MB)	14.45s (11468MB)
	ROH	10.95s (9319MB)	11.49s (9328MB)	10.42s (8294MB)	10.48s (8310MB)	79.53s (17931MB)	81.13s (18053MB)	29.11s (11849MB)	30.22s (11907MB)
	RCMH	1.99s (5351MB)	2.17s (5357MB)	0.80s (4451MB)	0.95s (4453MB)	6.42s (6529MB)	8.82s (6569MB)	4.68s (5020MB)	5.33s (5069MB)

TABLE VIII
LINEAR-NONLINEAR COMPETITIVE STRATEGY ANALYSIS UNDER DIFFERENT BIT LENGTHS. THE BEST PERFORMANCES ARE IN BOLDFACE

Task	Modality	IAPR TC-12				MIR-Flickr				NUS-WIDE				MS-COCO			
		16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits
$I \rightarrow T$	Image Only	0.5014	0.5202	0.5397	0.5481	0.7407	0.7522	0.7567	0.7652	0.6389	0.6570	0.6630	0.6686	0.8519	0.8718	0.8849	0.8978
	Text Only	0.5079	0.5270	0.5462	0.5583	0.7503	0.7659	0.7692	0.7795	0.6528	0.6650	0.6672	0.6759	0.8618	0.8860	0.8999	0.9039
	Linear Only	0.5038	0.5271	0.5456	0.5589	0.7456	0.7634	0.7698	0.7795	0.6451	0.6603	0.6631	0.6658	0.8548	0.8843	0.8978	0.9060
	Nonlinear Only	0.5011	0.5240	0.5420	0.5540	0.7425	0.7599	0.7686	0.7801	0.6419	0.6612	0.6680	0.6707	0.8520	0.8787	0.8970	0.9013
	Inversion	0.5001	0.5267	0.5393	0.5498	0.7431	0.7599	0.7688	0.7706	0.6461	0.6539	0.6644	0.6668	0.8526	0.8770	0.8924	0.8999
	RCMH	0.5113	0.5348	0.5506	0.5640	0.7568	0.7691	0.7755	0.7848	0.6573	0.6688	0.6781	0.6854	0.8679	0.8886	0.9031	0.9134
$T \rightarrow I$	Image Only	0.5914	0.6180	0.6386	0.6553	0.8166	0.8218	0.8294	0.8409	0.7402	0.7561	0.7686	0.7825	0.8893	0.9108	0.9262	0.9418
	Text Only	0.6021	0.6446	0.6760	0.7001	0.8289	0.8467	0.8542	0.8637	0.7765	0.7923	0.8023	0.8123	0.9029	0.9304	0.9480	0.9545
	Linear Only	0.5998	0.6436	0.6776	0.7016	0.8252	0.8433	0.8536	0.8603	0.7708	0.7983	0.8061	0.8108	0.8975	0.9294	0.9473	0.9552
	Nonlinear Only	0.5988	0.6417	0.6718	0.6979	0.8212	0.8419	0.8530	0.8620	0.7749	0.7929	0.8049	0.8110	0.8910	0.9226	0.9441	0.9523
	Inversion	0.5933	0.6390	0.6639	0.6880	0.8196	0.8364	0.8489	0.8518	0.7716	0.7912	0.8015	0.8077	0.8974	0.9252	0.9388	0.9496
	RCMH	0.6064	0.6514	0.6852	0.7083	0.8322	0.8488	0.8578	0.8666	0.7836	0.8013	0.8095	0.8218	0.9086	0.9341	0.9531	0.9637

performance when using different modalities. The competitive results of RCMH on various datasets were determined by fixing the final hash code length to 64 bits. Additionally, we reported RCMH results under various settings: ‘Image Only’, ‘Text Only’, ‘Linear Only’, ‘Nonlinear Only’, and ‘Inversion’. Specifically, ‘Image Only’ and ‘Text Only’ indicate that RCMH uses only image features or only text features, respectively, during the regeneration stage. ‘Linear Only’ and ‘Nonlinear Only’ denote that RCMH employs solely the linear generation module (6) and the nonlinear generation module (8), respectively. In contrast, the ‘Inversion’ setting swaps the roles: linear generation is applied to text features, while nonlinear generation processes image features.

The experimental results are presented in Fig. 11 and Table VIII. As shown in Fig. 11, the proportion of selected text

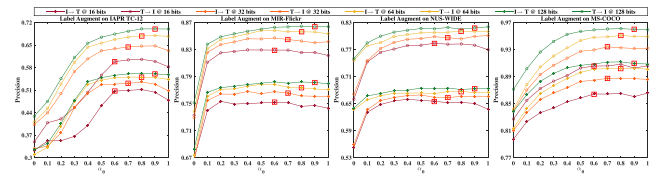


Fig. 13. The effect of α_0 in all bits. The best results are indicated by red rectangle.

modality is consistently larger than that of the image modality under the competitive strategy. This suggests that nonlinear text features are closer to the target real-valued space than linear image features, leading the model to favor text features in most sample pairs. Table VIII demonstrates that using only text features yields significantly higher MAP scores than using only

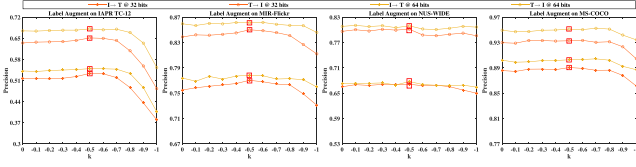


Fig. 14. The effect of negative parameter k in 32 and 64 bits. The best results are indicated by red rectangle.

image features in the regenerated term. Furthermore, RCMH outperforms other variants with solely linear or nonlinear in terms of MAP, i.e., ‘Linear Only’ and ‘Nonlinear Only’. Importantly, ‘Inversion’ shows the worst performance than ‘Linear Only’ and ‘Nonlinear Only’, confirming that image features are better suited for linear generation, while text features benefit more from nonlinear modeling. In summary, the evidence from Fig. 11 and Table VIII supports the conclusion that the competitive strategy dynamically selects the appropriate representation for hash code regeneration, such as linear image features or nonlinear text features, thereby achieving superior performance.

I. Convergence Analysis

As described in Section IV-G, RCMH demonstrates high efficiency in handling cross-modal retrieval tasks without the need for iterations. To further analyze its convergence and MAP results with the iterations T , experiments were conducted. Since RCMH is built upon the existing hash codes of FCMH, the SMF (2) was employed to represent the final loss values of the hash codes for both methods, thereby eliminating the impact of different objective functions. Specifically, the loss values of FCMH and RCMH at the i -th iteration are denoted as $LOSS_F^{(i)}$ and $LOSS_R^{(i)}$, respectively. To normalize these loss values, they are divided by the initial loss value of FCMH, $LOSS_F^{(1)}$, resulting in normalized losses defined as: $Loss_F^{(i)} = \frac{LOSS_F^{(i)}}{LOSS_F^{(1)}}$ and $Loss_R^{(i)} = \frac{LOSS_R^{(i)}}{LOSS_F^{(1)}}$. The normalized loss values $Loss_F^{(i)}$ and $Loss_R^{(i)}$ are obtained by fixing the hash code length to 64 bits. The curves of these normalized losses and the corresponding MAP results are illustrated in Fig. 12.

From Fig. 12, it is evident that RCMH significantly reduces the loss value of FCMH and rapidly enhances the MAP retrieval results. Notably, increasing the number of iterations in RCMH does not substantially decrease the loss value $Loss_R$ or improve the MAP results. This indicates that setting the iteration number T of RCMH to 1 achieves the best MAP results with minimal training cost. Thus, RCMH demonstrates its ability to converge quickly without requiring iterative optimizations.

J. Label Augmentation Analysis

This subsection focuses on analyzing the label augmentation term in RCMH, particularly the effectiveness of α_m , the negative parameter k and comparisons with recent SOTA label enhancement or label distribution methods. To achieve this, we first fixed all unrelated parameters and varied α_0 within the range $[0 : 1]$ and k within the range $[0 : -1]$ determine the optimal values of

α_m and k . Subsequently, the results were compared with LDL-LCLR (IJCAI₂₀₁₉) [31], GLE (TKDE₂₀₂₁) [38], LDL-LDM (TNNLS₂₀₂₃) [39] and LADH (ACMTKDD₂₀₂₄) [40].

As illustrated in Fig. 13, the retrieval results initially improve and then slightly deteriorate as α_0 increases. This trend can be attributed to the growing influence of label information on label augmentation with increasing α_0 . It is important to note that the optimal α_0 varies with different bit lengths, such as $\alpha_0 = 0.6$ on 16 bits and $\alpha_0 = 0.9$ on 128 bits. To adapt in these variations, we set $\hat{\alpha}$ as the initial parameter and define $\alpha_0 = \hat{\alpha} + (\sqrt{\frac{r}{16}} - 1) * 0.1$ for different bit lengths. Consequently, $\hat{\alpha}$ is set to 0.6 for all datasets to achieve optimal performance. Regarding the negative parameter k , Fig. 14 shows that the MAP results first improve and then decline as k decreases. This is because a large k fails to capture the negative information in $\mathbf{F}$, while an excessively small k can distort the positive label ‘1’ information. Therefore, k is set as -0.5 for all datasets to ensure optimal performance.

Given that LDL-LCLR, GLE, and LDL-LDM are primarily focused on handling single-modal features to enhance the label matrix, but are not specifically designed for cross-modal retrievals. For a fair comparison, we implemented all methods with features from one modality and recorded their training time, including the cross-modal label enhancement method LADH. The output of label distributions of all methods is then used as input for RCMH to obtain their Mean_MAP ((Image-to-Text MAP + Text-to-Image MAP)/2) results for 32-bit tasks. For instance, ‘Image’ in LDL-LCLR means that it generates label distributions with only the image features and original labels on IAPR TC-12. These learned label distributions from one modality are then treated as input labels in RCMH to obtain retrieval results (i.e., Image Mean_MAP: 0.3049), ensuring that only the input labels differ among all methods. In the experiment, a small training set (4,000 instances) was randomly selected from all datasets, considering the high computational complexity of some label enhancement methods. As depicted in Table IX, RCMH achieves the best Mean_MAP with the least time. The main reason may be that the novel label similarity augmentation term adopted by RCMH can not only balance information of different modalities but also preserve the original label structure in augmented labels; its effectiveness can be further referred to in Section IV-K. However, LDL-LCLR, GLE, and LDL-LDM may focus more on reconstructing missed or noisy labels, which may result in their relatively poor performance on retrieval tasks.

K. Ablation Analysis

To conduct a comprehensive analysis of RCMH, we performed ablation experiments on four datasets by removing specific terms from the original RCMH model. The results of these experiments, conducted with various code lengths, are presented in Table X. The symbol ‘*’ denotes the removal of a specific term.

In the ‘ $Loss_1*$ ’ experiment, we mitigated the influence of the label augmentation term from RCMH by setting $\mathbf{E} = \mathbf{G}$ in (12). The results in Table X demonstrate that the label augmentation term can improve retrieval performance by approximately 1% in terms of MAP and TOP@100, particularly for long hash-bits.

modality to 1500. The observations indicate that the performance of ‘ $Loss_4^*$ ’ is inferior to RCMH, suggesting that the nonlinear term in $Loss_4$ can preserve more nonlinear features than kernel regression methods.

‘ $Loss_5^*$ ’ refers to the removal of the competitive strategy in $Loss_5$ of RCMH, which swaps ‘0’ and ‘1’ in (9) to eliminate the influence of $Loss_5$. From Table X, we can observe that the MAP results in ‘ $Loss_5^*$ ’ are worse than RCMH. It indicates that modality feature selection is important for preserving modality-specific properties in the final hash codes and obtaining better retrieval results.

‘ $Loss_6^*$ ’ involves removing the regenerated term of (11) from the objective function (12), meaning that $\mathbf{Z}$ is selected as the final hash codes instead of $\mathbf{B}$. As shown in Table X, ‘ $Loss_6^*$ ’ contributes to 10 – 20% improvements in retrieval performance in terms of MAP and TOP@100 for RCMH. This demonstrates that the regenerated strategy is crucial for RCMH and significantly enhances retrieval performance.

L. Parameter Setting and Sensitivity Analysis

To demonstrate the efficacy of all parameters in RCMH, we performed a series of experiments to analyze their sensitivity, focusing on μ and γ . Given that θ is determined by the dataset itself, we set $\theta = 10$ for all datasets to avoid over-fitting in hash function learning. Specifically, we plotted the MAP results of RCMH on all datasets by varying μ and γ within the range of $[0, 1e^6]$ and $[5e^{-4}, 1]$ respectively, as shown in Fig. 15.

The parameter μ facilitates the information preservation of the existing hash codes $\mathbf{U}$ into the final hash codes $\mathbf{B}$. Although the performance improvement is not immediately apparent, the MAP results improve with an increase in μ . It slightly improves the quality of the final hash codes by preserving the information of the existing hash codes and bridging between the existing hash codes and the final hash codes. However, RCMH’s performance deteriorates when $\mu > 1e^3$. In this study, we set $\mu = 1e^2$ for all datasets to achieve the best MAP results.

The parameter γ dictates the learning rate of RCMH, plays a crucial role in ensuring convergence, and in achieving optimal retrieval results in RCMH. Fig. 15 shows that the MAP results improve with an increase of γ . However, when γ becomes excessively large, the MAP results begin to decrease. In addition, the best parameters for γ vary across different datasets. The reason may be related to the differences in the size and the representations of the different datasets. As shown in Fig. 15, we set $\gamma = 5e^{-2}$ on MIR-Flickr and $\gamma = 5e^{-3}$ on other datasets to obtain the best results.

In conclusion, while the parameters indeed influence the retrieval results, RCMH can obtain relatively robust performance within a certain parameter range.

M. Comparison With Deep Hashing

Deep learning methods have recently demonstrated outstanding performance in cross-modal hashing retrieval, owing to their ability to extract highly nonlinear features from data and learn more powerful regression functions [77], [78]. To demonstrate that RCMH can outperform deep cross-modal hashing methods despite using a shallow hashing model, we conducted

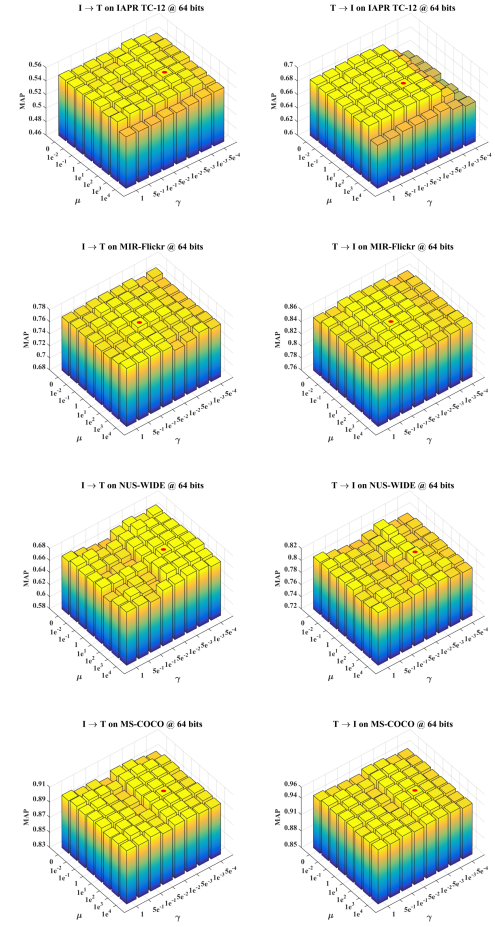


Fig. 15. MAP versus parameter μ and γ , the best results are indicated by red dots.

experiments comparing it against SOTA deep approaches based on different neural network architectures, such as very deep convolutional networks (VGG) [79] and contrastive language-image pre-training (CLIP) [80]. The VGG-based methods include DCMH (CVPR₂₀₁₇) [57], DCUCH (TKDE₂₀₂₂) [58], Bi-CMR (AAAI₂₀₂₂) [60], BSTH (SIGIR₂₀₂₃) [59] and SCH (TPAMI₂₀₂₄) [62]. The CLIP-based methods include UMoED (ToMM₂₀₂₅) [63], DIMCH (TIP₂₀₂₅) [64] and DDBH (TCSVT₂₀₂₅) [65]. Following the VGG setting, the image features are extracted as 4,096-dimensional features by a CNN deep network on ImageNet, the text features are extracted by bidirectional encoder representations from transformers (BERT) on IAPR TC-12 [81], and extracted by BOW vectors on the other three datasets in the experiments. For the CLIP setting, we employed the CLIP (ViT-B/32) pre-trained model, where the image encoder is a Vision Transformer and the text encoder follows a GPT-2 architecture, producing 512-dimensional features for each modality [82].

All experimental results of VGG and CLIP structures are presented in Table XI. Here, RCMH_{vgg} denotes the variant of RCMH that uses features extracted by the VGG architecture, and RCMH_{clip} denotes the variant using features extracted from the CLIP architecture. RCMH outperforms VGG-based methods using VGG features and achieves better MAP results than CLIP-based methods using CLIP features. This advantage

stems from RCMH's ability to enhance the initial hash codes produced by the SMF model into final hash codes through its linear-nonlinear competitive strategy. Specifically, the regenerated term refines and generates better final hash codes with these linear and nonlinear features from different modalities. However, the performances of SCH in $T \rightarrow I$ tasks on MIR-Flickr exceed $RCMH_{vgg}$ under low-bit tasks; this may be attributed to the fact that SCH employs Hamming space to reduce subspace gaps and uses a semantic channel to mitigate loss function oscillation. Similarly, DIMCH achieves some better results than $RCMH_{clip}$ on MS-COCO, as it employs a weighted cosine triplet loss to better exploit the rich supervised information provided by the dataset's complex semantic labels. Nevertheless, RCMH generally outperforms both SCH and DIMCH across all datasets on two deep learning structures.

These experiments demonstrate that RCMH can be adapted into a deep hashing framework by simply plugging in features extracted from deep neural networks, while still achieving SOTA cross-modal retrieval performance. Although RCMH is not an end-to-end deep model, it outperforms existing deep SOTA methods and offers greater controllability and interpretability. Moreover, it can serve as a plug-in embedding module to further enhance the performance of deep learning based SOTA methods.

V. CONCLUSION

This paper presents a novel supervised hashing framework, termed Regenerated Cross-Modal Hashing (RCMH), designed to further improve the performance of the existing methods with high efficiency. Firstly, RCMH proposes a simple yet effective label augmentation technique to augment the label information. Subsequently, a linear-nonlinear reconstruction is proposed to capture the linear features from image data and the nonlinear features from text data. In addition, a competitive strategy is proposed to identify the impact of features in generating the reconstruction binary space. Finally, the final hash codes are regenerated from the reconstruction binary space by the regenerated hashing term. Experimental results on four widely used datasets demonstrate that RCMH outperforms several SOTA shallow and deep hashing techniques. Furthermore, RCMH can also be applied to SOTA shallow hashing methods, resulting in performance improvements and bit length conversion. The general design of RCMH enables it to enhance not only shallow hashing methods but also deep hashing approaches. As a model post-processing framework, RCMH can leverage the rich high-level features and hash codes produced by deep networks without requiring modifications. In future work, we will investigate the integration of RCMH with advanced asymmetric deep hashing frameworks, aiming to capture their modality-specific representations through our regenerated mechanism to achieve better results.

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